

LCFC Confidential

GS454/GS554(NMD031) MB Schematics Document

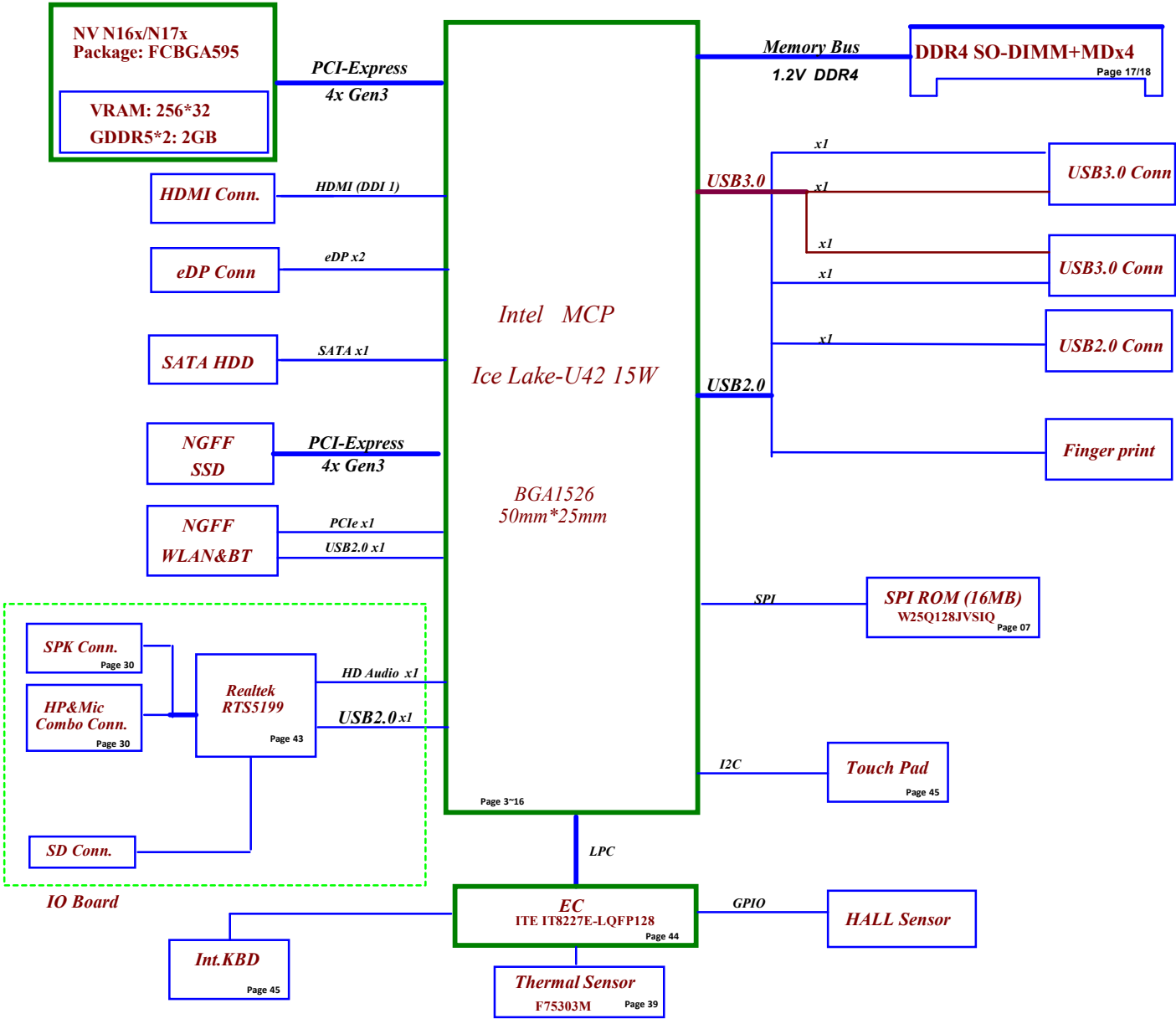
ICL U42 with DDR4 + Nvidia N16V-GM

2019-10

REV: 0.1

Security Classification	LC Future Center Secret Data			Title	
Issued Date	2015/08/20	Deciphered Date	2016/08/20	Cover Page	
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Voltage Rails (O --> Means ON , X --> Means OFF)

Power Plane / State	V9B+	+3VALW +5VALW +3VALW_PCH +1.8VALW VCC_AUX	+1.2V +2.5V_DDR +VCCST +VCCSTG	+5VS +3VS +1.8VS +CPU_CORE +0.6VS
S0	O	O	O	O
S3	O	O	O	X
S3 Battery only	O	O	O	X
S5 S4 AC Only	O	O	X	X
S5 S4 Battery only	O	X	X	X
S5 S4 AC & Battery don't exist	X	X	X	X

STATE	SIGNAL	SLP_S3#	SLP_S4#	SLP_S5#	+VALW	+V	+VS	Clock
Full ON		HIGH	HIGH	HIGH	ON	ON	ON	ON
S3 (Suspend to RAM)		LOW	HIGH	HIGH	ON	ON	OFF	OFF
S4 (Suspend to Disk)		LOW	LOW	LOW	ON	OFF	OFF	OFF
S5 (Soft OFF)		LOW	LOW	LOW	ON	OFF	OFF	OFF

HSIO PORT	Function
USB3.0	1 USB3.0 Conn
	2 USB3.0 Conn
	3 NC
	4 NC
	5
	6
USB2.0	1 USB3.0 Conn
	2 USB3.0 Conn
	3 NC
	4 NC
	5 Camera
	6 Touch Screen
	7 Finger Print
	8 Card Reader
	9 USB2.0 conn
	10 Bluetooth
PCIE	5~8 X4 DGPU
	9 WLAN
	10 NC
	11 SATA HDD
	12 NC
	13~16 X4 PCIE/SATA SSD

BOM Structure	BTO Item
@	Un-stuff
14@	For 14" part
15@	For 15" part
CD@	For cost down
EMC@	For EMC part
EMC_15@	For EMC 15" part
EMC_NS@	For EMC un-stuff part
ME@	For ME part
UMA@	For UMA part
OPT@	For NV GPU part
OPTN16@	For NV N16S-GTR GPU part
OPTN17@	For NV N17S-G1 GPU part
TS@	For touch screen part
TP@	For Touch Pad Part

SMBUS Control Table

	SOURCE	BATT	Charger	DGPU	IT8227E	Memory Down	PCH	PMIC	SODIMM	Thermal Sensor	WLAN W1MAX
EC_SMB_CK1 EC_SMB_DA1	IT8227E +3VL_EC	V	V	X	V +3VL_EC	X	X	X	X	X	X
EC_SMB_CK0 EC_SMB_DA0	IT8227E +3VS	X	X	V +3VG_AON	V +3VS	X	X	X	X	V	X
EC_SMB_CK3 EC_SMB_DA3	IT8227E +3VAWL	X	X	X	V	X	X	V	X	X	X
PCH_SMB_CLK PCH_SMB_DATA	PCH +3VALW_PCH	X	X	X	X	X	V +3VALW_PCH	X	V +3VS	X	X

EC SMBus1 address

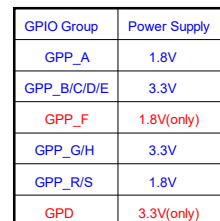
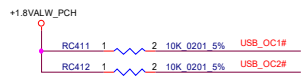
EC SMBus2 address

EC SMBus3 address

PCH SM Bus address

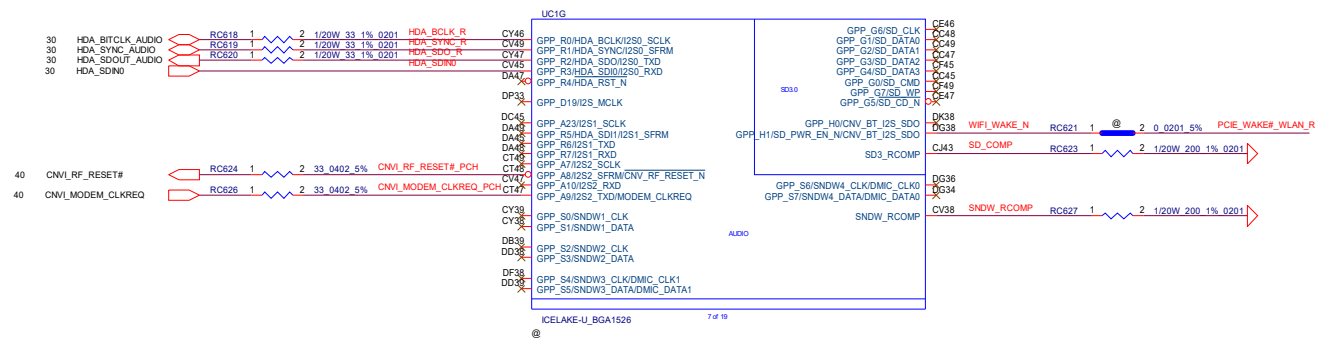
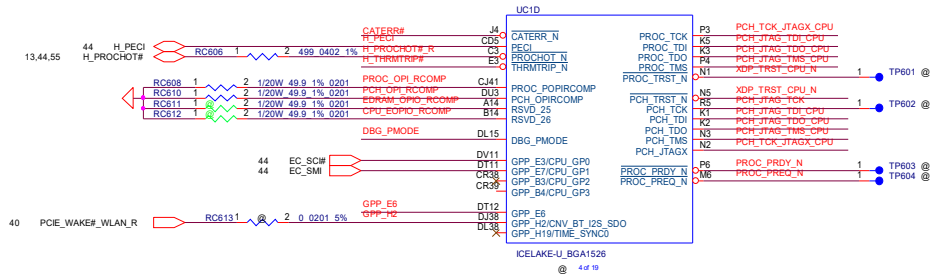
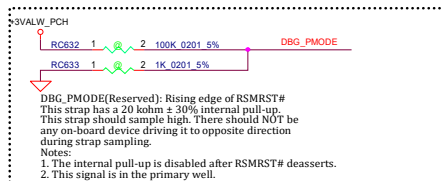
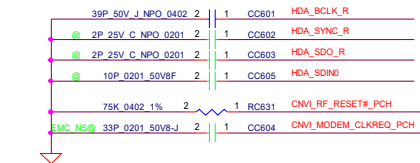
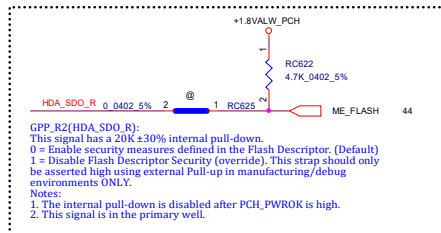
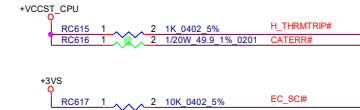
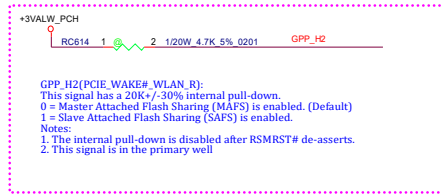
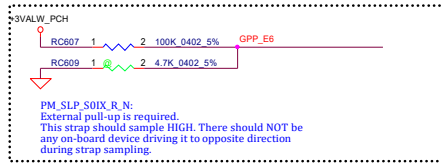
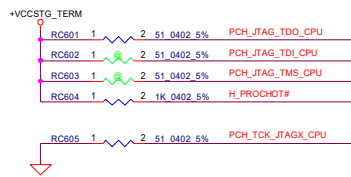
Device	Address	Device	Address	Device	Address	Device	Address
Smart Battery	need to update	Thermal Sensor(NCT7718W)	1001_100xb	PMIC	need to update	DDR4 SODIMM	need to update
Charger	0001 0010 b	DGPU	need to update				

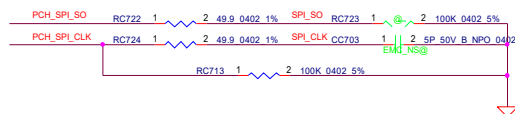
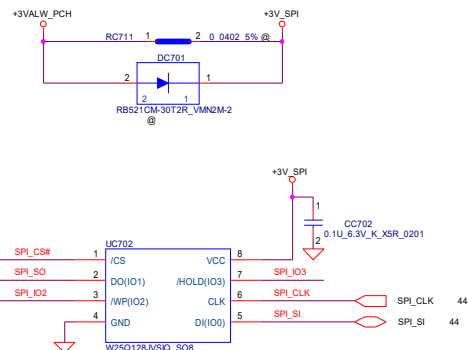
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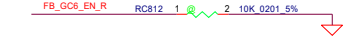
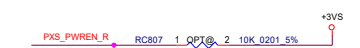
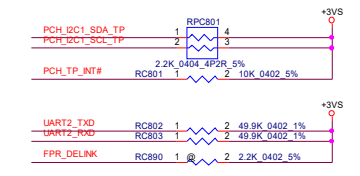
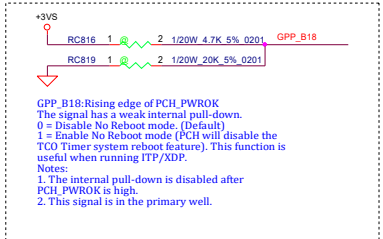
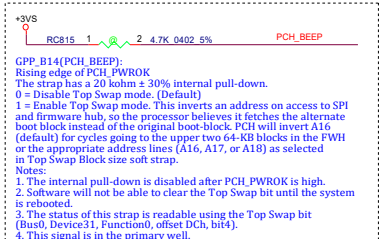
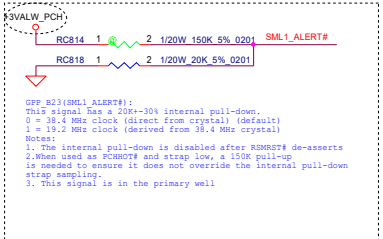
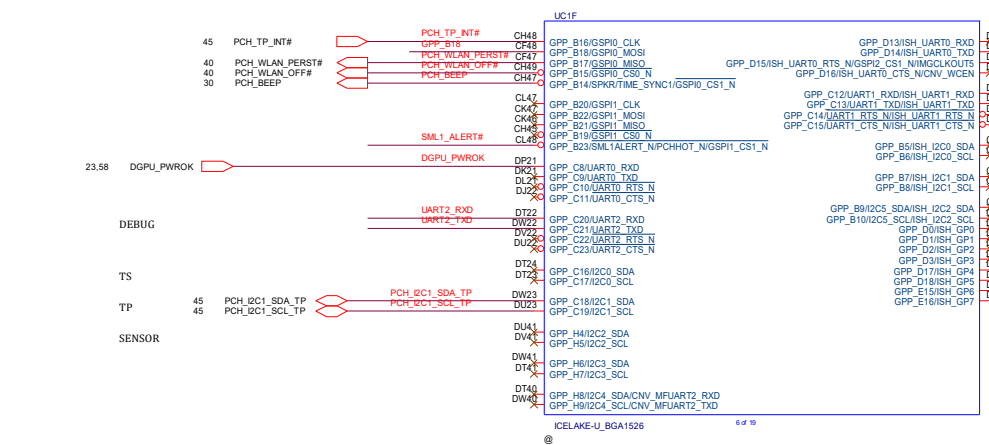
GPIO Group	Power Supply
GPP_A	1.8V
GPP_B/C/D/E	3.3V
GPP_F	1.8V(only)
GPP_G/H	3.3V
GPP_R/S	1.8V
GPD	3.3V(only)







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20 PCIE_CRX_GTX_P15_8]

20 PCIE_CTX_C_GRX_N5_8]

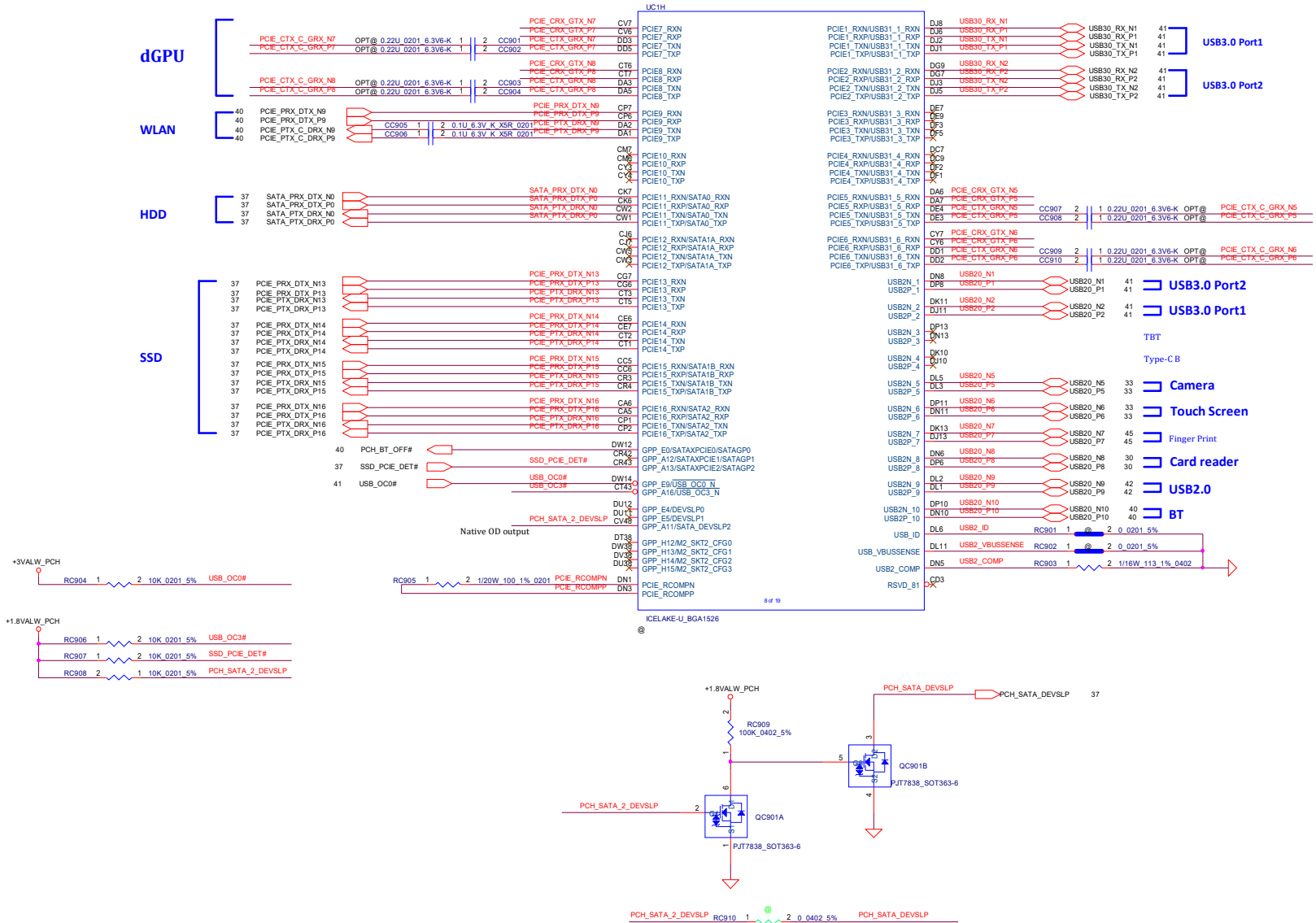
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dGPU

WLAN

HDD

SSD

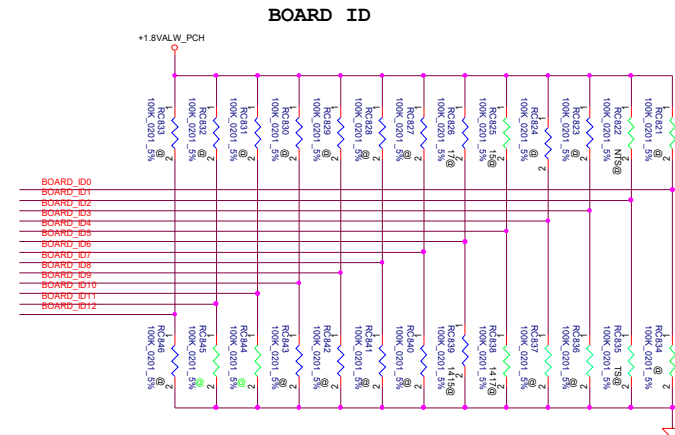
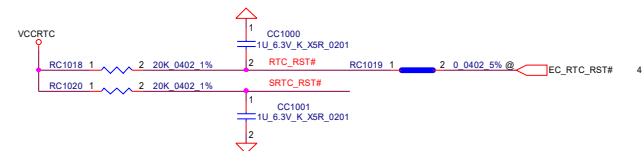
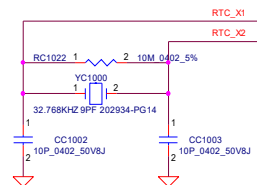
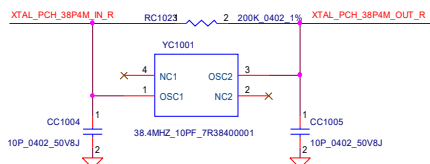
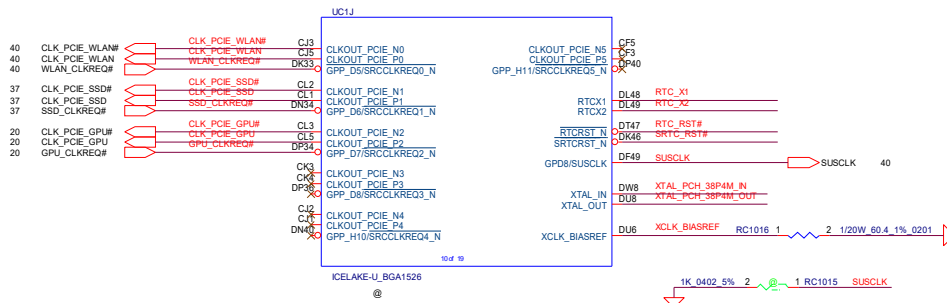
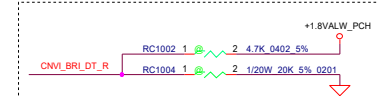


1. DEVS LP is an open-drain pin on the PCH side and is not required external pull-up or pull-down. The PCH will tri-state this pin to signal to the SATA device that it may enter a lower power state (pin will go high due to pull-up that's internal to the SATA device, per DEVS LP specification). PCH will drive pin low to signal an exit from DEVS LP state.
2. DEVS LP is supported in direct connect, mSATA/mPCIe, uSATA, M.2.
3. 1 DEVS LP pin is required to support EACH DEVS LP enabled RAID storage device.
(Example: 2 DEVS LP pins are required to support 2 DEVS LP RAID storage devices).

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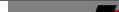


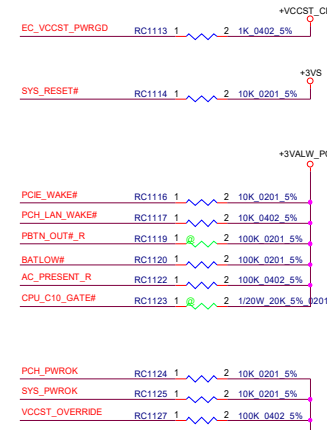
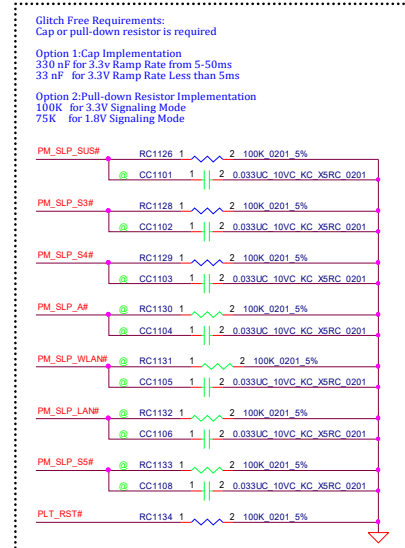
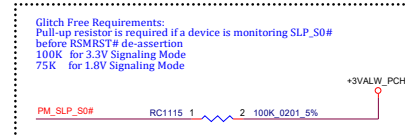
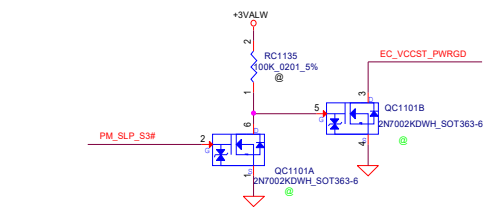
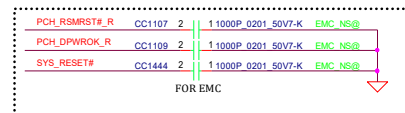
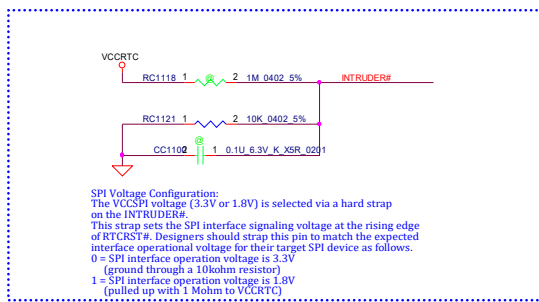
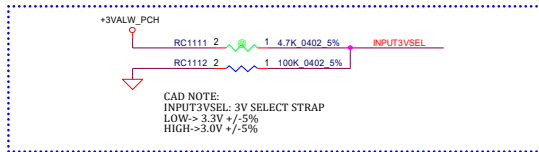
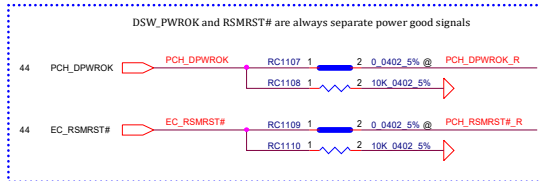
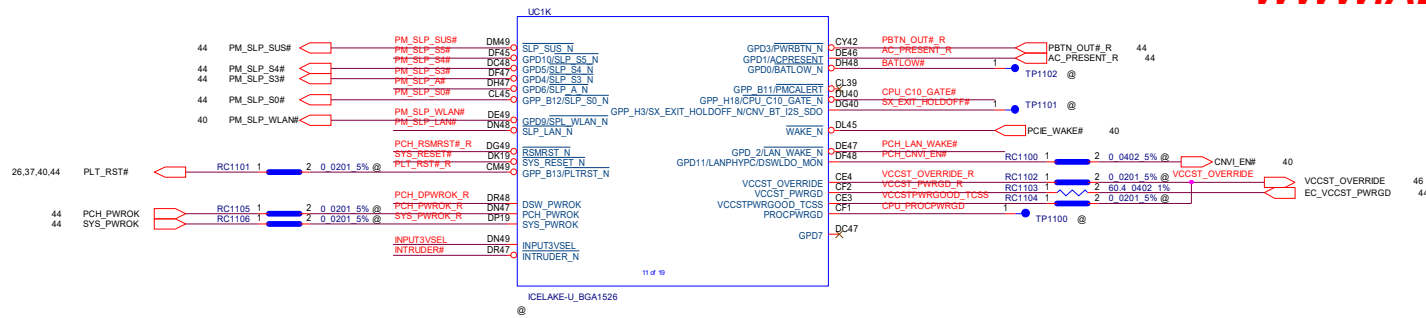
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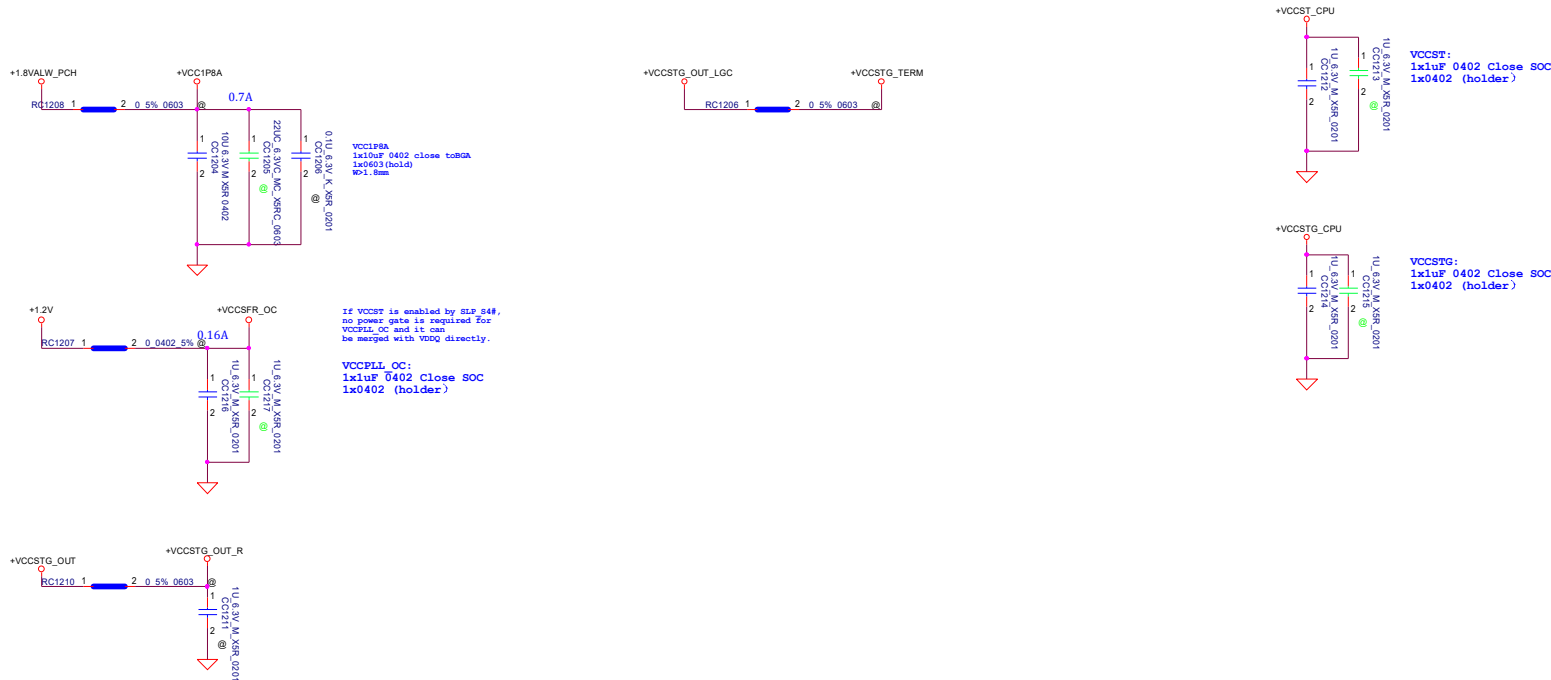
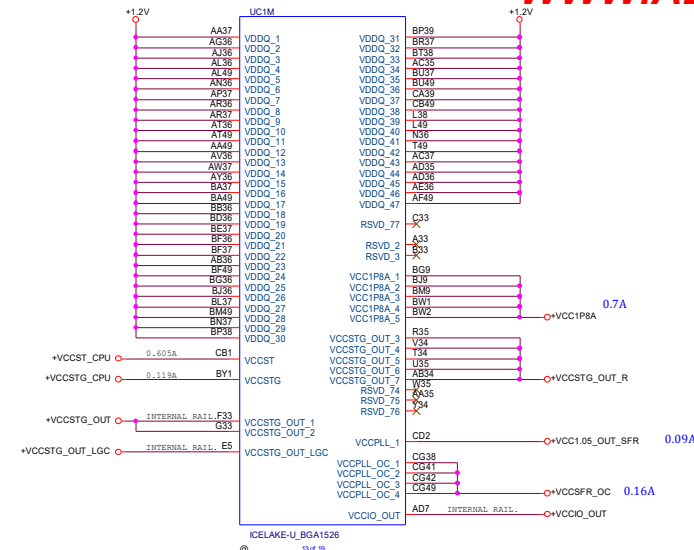
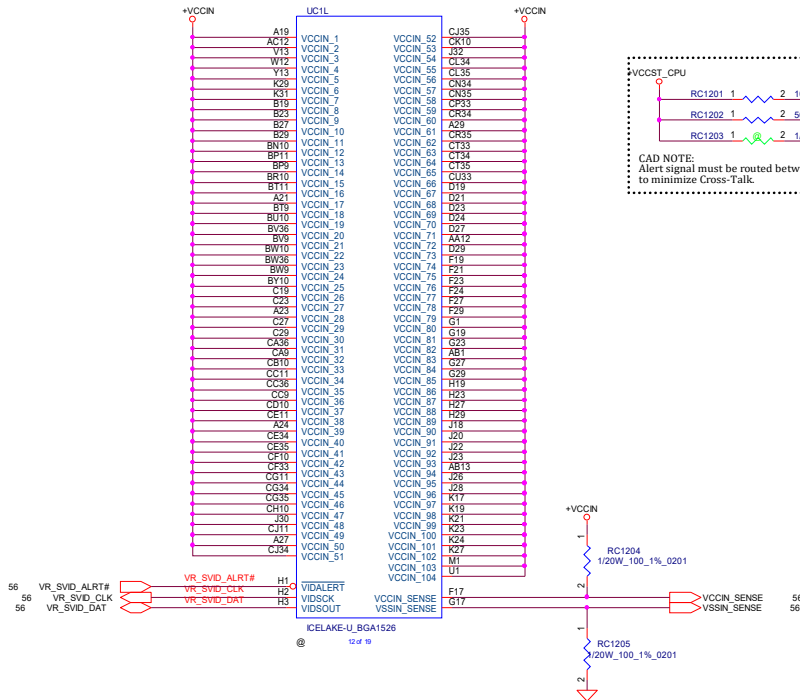
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BOARD_ID67/8/9	Samsung	0000
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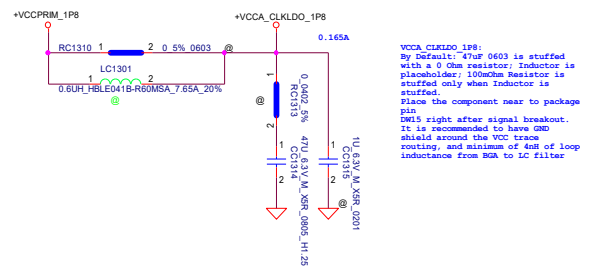
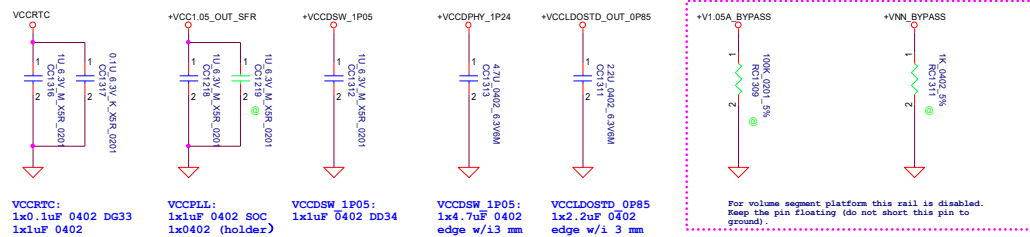
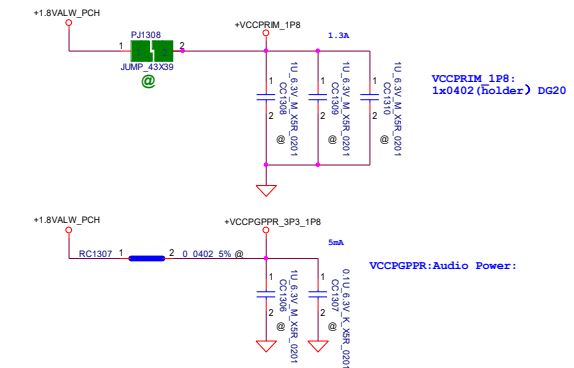
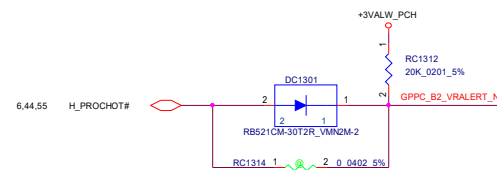
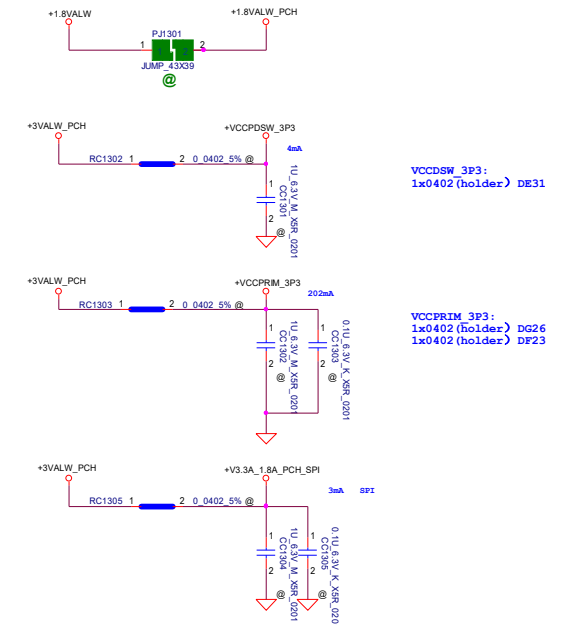
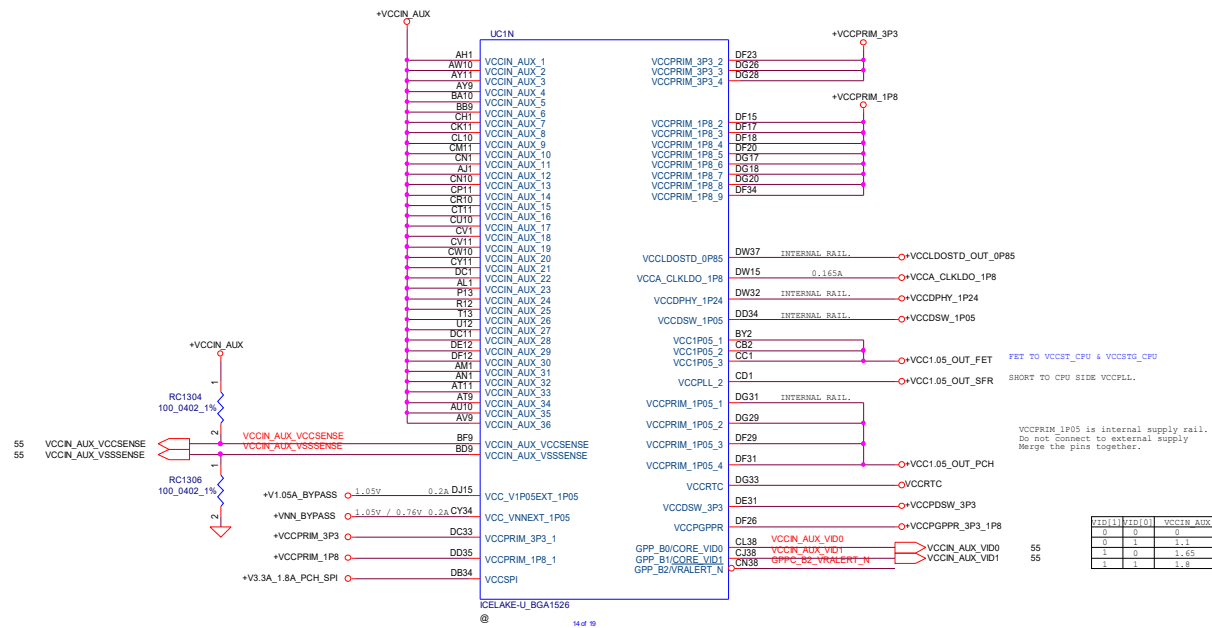
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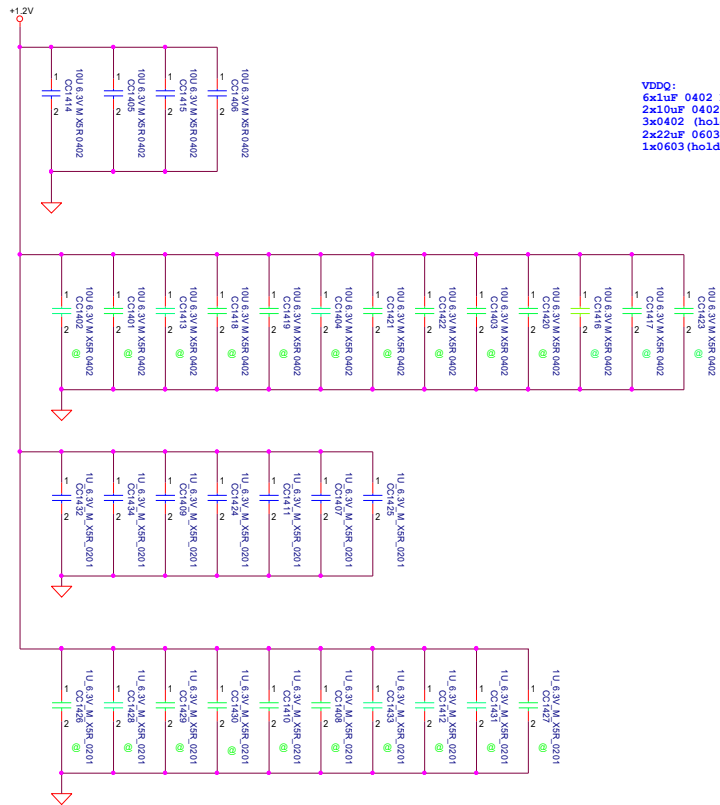
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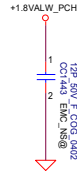
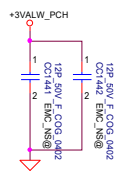
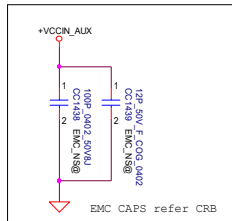
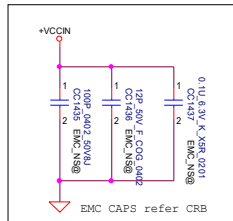
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VDDQ:
6x1uF 0402 Back ,Outer row
2x10uF 0402 back Outer row
3x0402 (holder) back Outer row
2x22uF 0603 (SODIMM need)
1x0603 (holder)



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UC1O		
A11	VSS 1	VSS 75
A46	VSS 2	VSS 76
BA45	VSS 3	VSS 77
BA47	VSS 4	VSS 78
BB11	VSS 5	VSS 79
BB3	VSS 6	VSS 80
BB7	VSS 7	VSS 81
BC37	VSS 8	VSS 82
BD3	VSS 9	VSS 83
BD38	VSS 10	VSS 84
BD39	VSS 11	VSS 85
BD41	VSS 12	VSS 86
BD42	VSS 13	VSS 87
BD43	VSS 14	VSS 88
BD45	VSS 15	VSS 89
BD49	VSS 16	VSS 90
BD6	VSS 17	VSS 91
BD6	VSS 18	VSS 92
BD7	VSS 19	VSS 93
BD7	VSS 20	VSS 94
BE2	VSS 21	VSS 95
BE2	VSS 22	VSS 96
BE3	VSS 23	VSS 97
BE45	VSS 24	VSS 98
BE47	VSS 25	VSS 99
BE7	VSS 26	VSS 100
BC3	VSS 27	VSS 101
BC41	VSS 28	VSS 102
BC7	VSS 29	VSS 103
BC37	VSS 30	VSS 104
BJ1	VSS 31	VSS 105
BJ2	VSS 32	VSS 106
BJ3	VSS 33	VSS 107
BJ4	VSS 34	VSS 108
BJ45	VSS 35	VSS 109
BJ47	VSS 36	VSS 110
BJ49	VSS 37	VSS 111
BJ49	VSS 38	VSS 112
BJ7	VSS 39	VSS 113
BJ7	VSS 40	VSS 114
BM11	VSS 41	VSS 115
BM3	VSS 42	VSS 116
BM45	VSS 43	VSS 117
BM47	VSS 44	VSS 118
BM6	VSS 45	VSS 119
AA47	VSS 46	VSS 120
BM7	VSS 47	VSS 121
BP1	VSS 48	VSS 122
BP2	VSS 49	VSS 123
BP3	VSS 50	VSS 124
BP43	VSS 51	VSS 125
BP7	VSS 52	VSS 126
BR45	VSS 53	VSS 127
BR49	VSS 54	VSS 128
AB11	VSS 55	VSS 129
AB3	VSS 56	VSS 130
AB38	VSS 57	VSS 131
AB39	VSS 58	VSS 132
AB41	VSS 59	VSS 133
AB42	VSS 60	VSS 134
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AB5	VSS 63	VSS 137
AB6	VSS 64	VSS 138
AC45	VSS 65	VSS 139
AD49	VSS 66	VSS 140
AD10	VSS 67	VSS 141
AD11	VSS 68	VSS 142
AD34	VSS 69	VSS 143
AD37	VSS 70	VSS 144
A3	VSS 71	VSS 145
AE1	VSS 72	VSS 146
AF37	VSS 73	VSS 147
VSS 74	VSS 74	VSS 148

Net Pin
ICELAKE-U, BGA1526
@

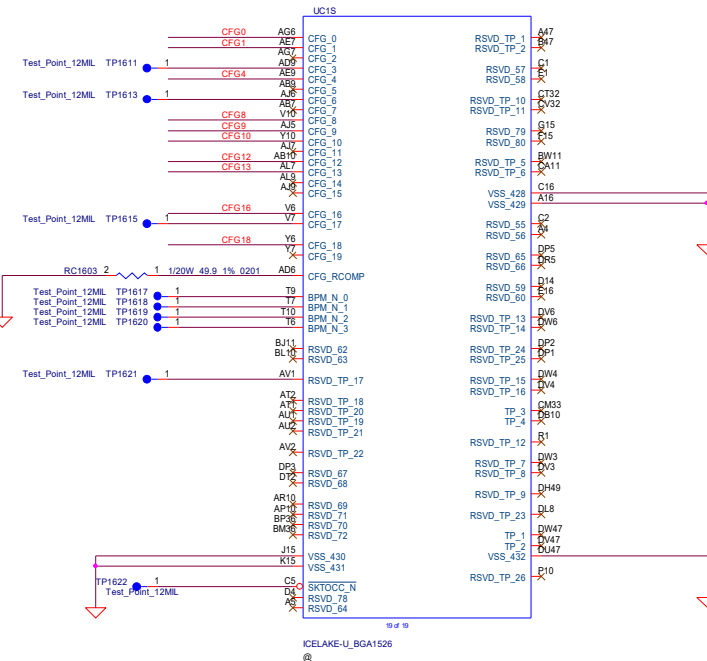
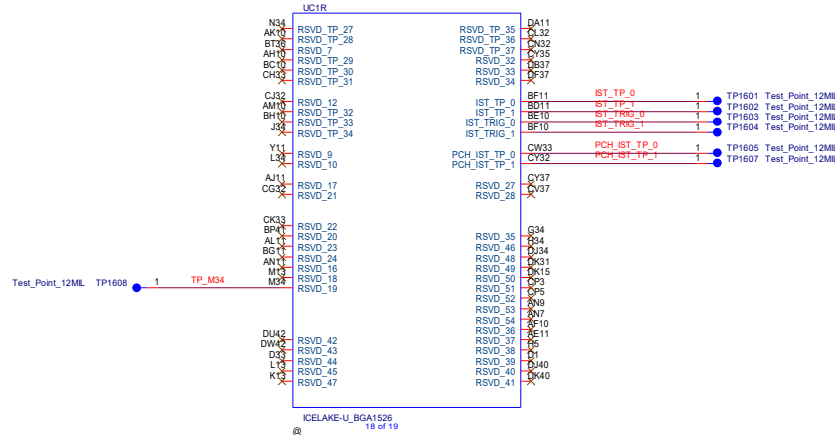
UC1P		
BT3	VSS 149	VSS 223
BT39	VSS 150	VSS 224
BT41	VSS 151	VSS 225
BT42	VSS 152	VSS 226
BT43	VSS 153	VSS 227
BT7	VSS 154	VSS 228
BU45	VSS 155	VSS 229
BU47	VSS 156	VSS 230
BU11	VSS 157	VSS 231
BU2	VSS 158	VSS 232
BU3	VSS 159	VSS 233
BU7	VSS 160	VSS 234
BU3	VSS 161	VSS 235
BU37	VSS 162	VSS 236
BU45	VSS 163	VSS 237
BU6	VSS 164	VSS 238
BU7	VSS 165	VSS 239
BU37	VSS 166	VSS 240
BU45	VSS 167	VSS 241
BU49	VSS 168	VSS 242
C11	VSS 169	VSS 243
C13	VSS 170	VSS 244
C14	VSS 171	VSS 245
C17	VSS 172	VSS 246
C21	VSS 173	VSS 247
C24	VSS 174	VSS 248
C31	VSS 175	VSS 249
C34	VSS 176	VSS 250
C39	VSS 177	VSS 251
C48	VSS 178	VSS 252
C49	VSS 179	VSS 253
C6	VSS 180	VSS 254
CA3	VSS 181	VSS 255
CA38	VSS 182	VSS 256
CA41	VSS 183	VSS 257
CA42	VSS 184	VSS 258
CA43	VSS 185	VSS 259
CA7	VSS 186	VSS 260
CB37	VSS 187	VSS 261
CB45	VSS 188	VSS 262
CB7	VSS 189	VSS 263
CC3	VSS 190	VSS 264
CC7	VSS 191	VSS 265
CE37	VSS 192	VSS 266
CE45	VSS 193	VSS 267
CE49	VSS 194	VSS 268
CE9	VSS 195	VSS 269
CG37	VSS 196	VSS 270
CG39	VSS 197	VSS 271
CG43	VSS 198	VSS 272
CG45	VSS 199	VSS 273
CG47	VSS 200	VSS 274
CG9	VSS 201	VSS 275
CH3	VSS 202	VSS 276
CH5	VSS 203	VSS 277
CJ37	VSS 204	VSS 278
CJ42	VSS 205	VSS 279
CJ8	VSS 206	VSS 280
CK45	VSS 207	VSS 281
CK49	VSS 208	VSS 282
CK9	VSS 209	VSS 283
CL37	VSS 210	VSS 284
CL42	VSS 211	VSS 285
CL49	VSS 212	VSS 286
CM45	VSS 213	VSS 287
CM47	VSS 214	VSS 288
CM9	VSS 215	VSS 289
CN3	VSS 216	VSS 290
CN37	VSS 217	VSS 291
CN39	VSS 218	VSS 292
CN5	VSS 219	VSS 293
CP9	VSS 220	VSS 294
CR32	VSS 221	VSS 295
VSS 222	VSS 222	VSS 296

Net Pin
ICELAKE-U, BGA1526
@

UC1Q		
DJ33	VSS 297	VSS 362
DJ36	VSS 298	VSS 363
DJ42	VSS 299	VSS 364
DK3	VSS 300	VSS 365
DK4	VSS 301	VSS 366
DK43	VSS 302	VSS 367
DK8	VSS 303	VSS 368
DL10	VSS 304	VSS 369
DL13	VSS 305	VSS 370
DL44	VSS 306	VSS 371
DL47	VSS 307	VSS 372
DM47	VSS 308	VSS 373
DN15	VSS 309	VSS 374
DN19	VSS 310	VSS 375
DN24	VSS 311	VSS 376
DN31	VSS 312	VSS 377
DN36	VSS 313	VSS 378
DN42	VSS 314	VSS 379
DR15	VSS 315	VSS 380
DR49	VSS 316	VSS 381
DT1	VSS 317	VSS 382
DT10	VSS 318	VSS 383
DT15	VSS 319	VSS 384
DT20	VSS 320	VSS 385
DT27	VSS 321	VSS 386
DT3	VSS 322	VSS 387
DT32	VSS 323	VSS 388
DT37	VSS 324	VSS 389
DT42	VSS 325	VSS 390
DT49	VSS 326	VSS 391
DT6	VSS 327	VSS 392
DT7	VSS 328	VSS 393
DT8	VSS 329	VSS 394
DT1	VSS 330	VSS 395
DT10	VSS 331	VSS 396
DT15	VSS 332	VSS 397
DT2	VSS 333	VSS 398
DT20	VSS 334	VSS 399
DT27	VSS 335	VSS 400
DT32	VSS 336	VSS 401
DT37	VSS 337	VSS 402
DT48	VSS 338	VSS 403
DT49	VSS 339	VSS 404
DT7	VSS 340	VSS 405
DV2	VSS 341	VSS 406
DV44	VSS 342	VSS 407
DV48	VSS 343	VSS 408
DV5	VSS 344	VSS 409
DV1	VSS 345	VSS 410
DW10	VSS 346	VSS 411
DW2	VSS 347	VSS 412
DW20	VSS 348	VSS 413
DW27	VSS 349	VSS 414
DW44	VSS 350	VSS 415
DW48	VSS 351	VSS 416
DW49	VSS 352	VSS 417
DW49	VSS 353	VSS 418
DW7	VSS 354	VSS 419
E11	VSS 355	VSS 420
E34	VSS 356	VSS 421
E38	VSS 357	VSS 422
E39	VSS 358	VSS 423
E42	VSS 359	VSS 424
E6	VSS 360	VSS 425
VSS 361	VSS 361	VSS 426
VSS 427	VSS 427	VSS 427

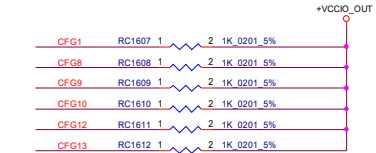
Net Pin
ICELAKE-U, BGA1526
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17 of 19





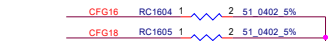
Stall reset sequence after PCU PLL lock until de-asserted

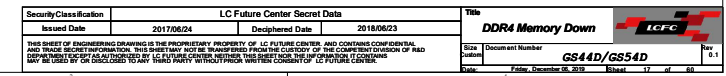
CFG0	1:Normal(Default) *
	0:Stall

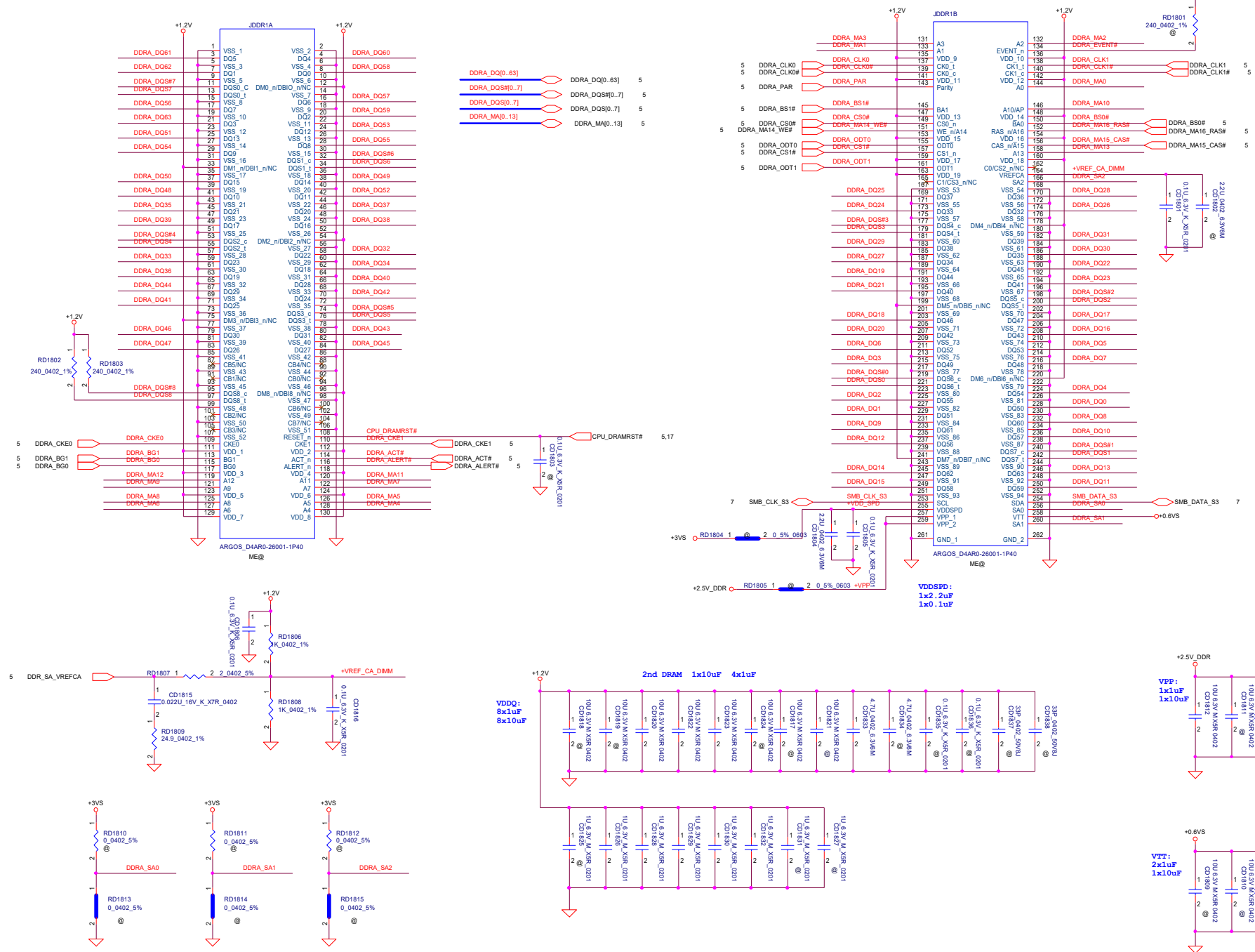


Embedded Display Port Presence Strap

CFG4	1:Disable
	0:Enable(Default) *







SPD Address = A0

N16x GPIO

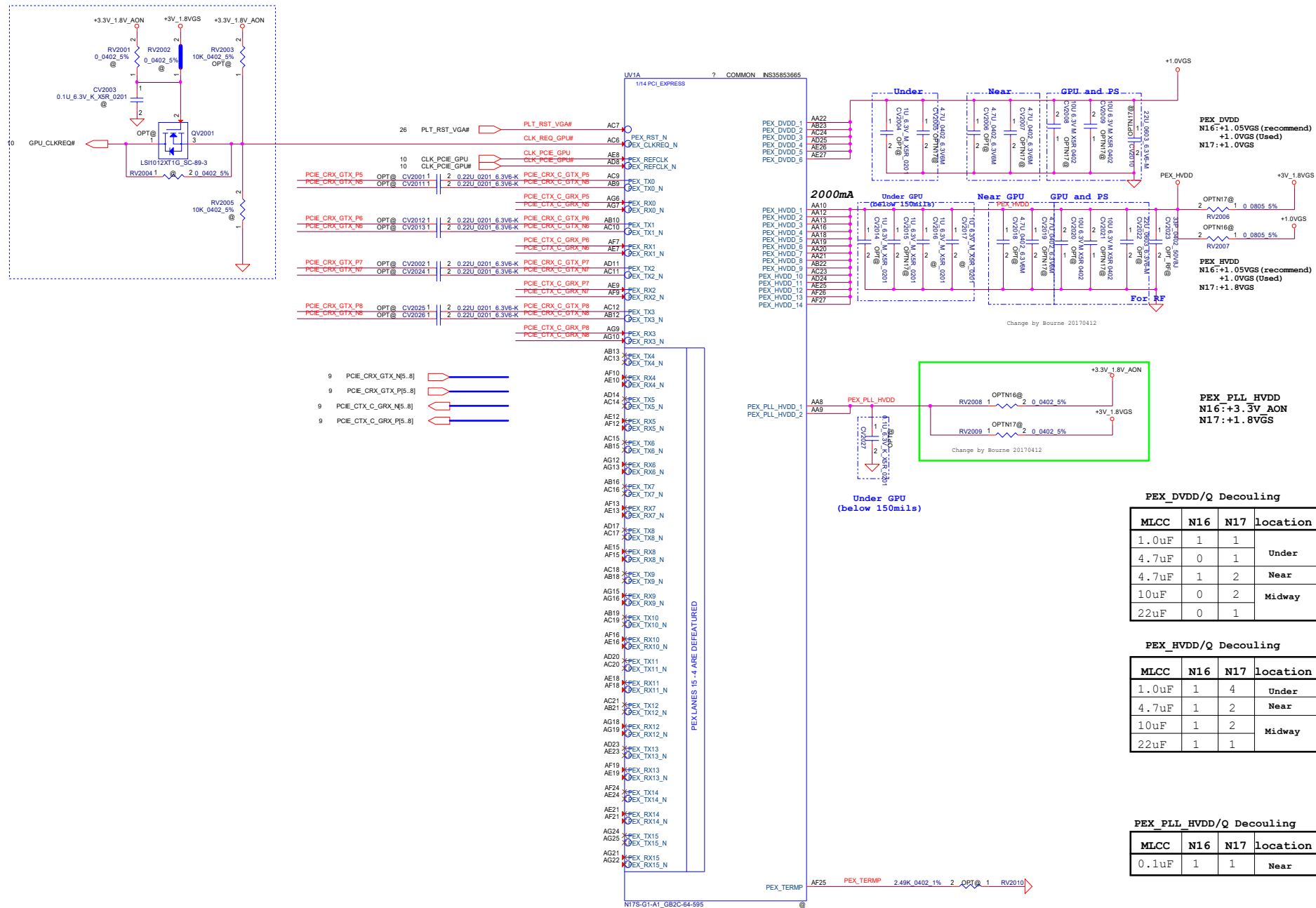
GPIO	I/O	ACTIVE	Function Description
GPIO0	OUT	-	GPU Core VDD PWM control signal
GPIO1	OUT	N/A	FB Enable for GC6 2.0
GPIO2	OUT	N/A	
GPIO3	OUT	N/A	
GPIO4	OUT	N/A	
GPIO5	OUT	N/A	GPU power sequencing---3V3_MAIN_EN
GPIO6	IN	-	GPU wake signal for GC6 2.0
GPIO7	OUT	N/A	
GPIO8	I/O	-	System side PCIe reset Monitor
GPIO9	I/O	N/A	2.2K Pull-up
GPIO10	OUT		FBVREF_ALTV for GDDR5
GPIO11	OUT	-	
GPIO12	IN		AC Power Detect Input (10K pull High)
GPIO13	OUT	-	Phase Shedding
GPIO14	IN	N/A	
GPIO15	IN	N/A	
GPIO16		N/A	
GPIO17	IN	N/A	
GPIO18	IN	N/A	
GPIO19	IN	N/A	
GPIO20		N/A	
GPIO21	OUT		GPU PCIe self-reset control
OVERT	OUT		Active Low Thermal Catastrophic Over Temperature

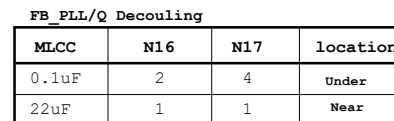
Performance Mode P0 TDP and EDP-Continuous current (GDDR5)

	GPU	Mem	Min Core Clk	NVVDD			FBVDD (1.35V)		FBVDDQ (GPU+Mem) (1.35V)		(1.05V) (6)		Other (3.3V)	
Products	(W)	(W)	(MHz)	(V)	(A)	(W)	(A)	(W)	(A)	(W)	(mA)	(W)	(mA)	(W)
N16S-GMR	16	1.6	849	TBD	19	TBD	2	TBD	4.2	TBD	800	TBD	60	TBD
N16S-GTR	18	1.7	967		26.5		2		4.2		800		60	

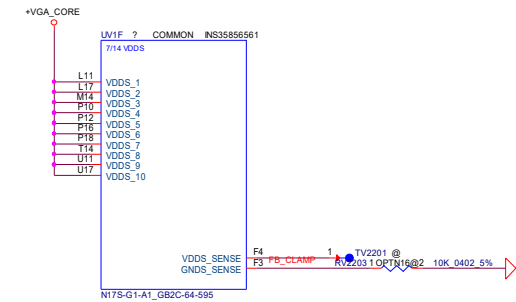
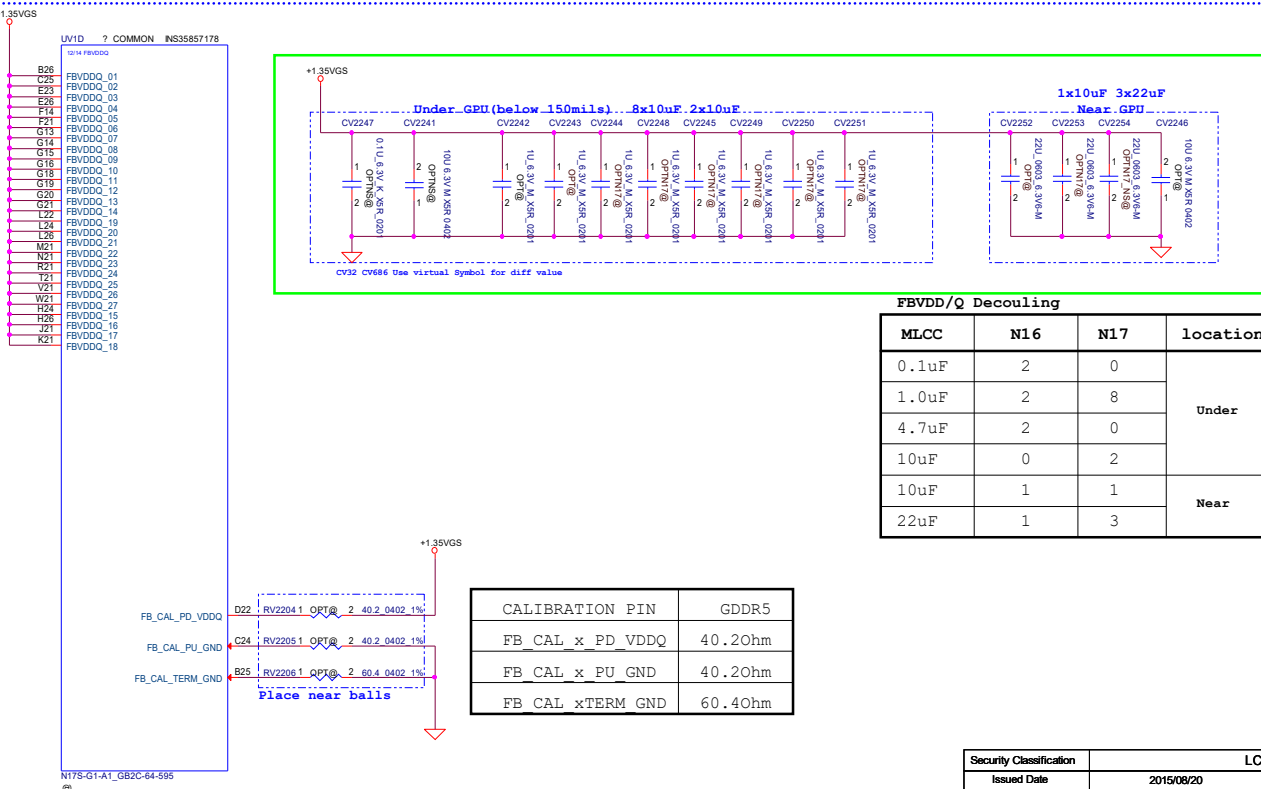
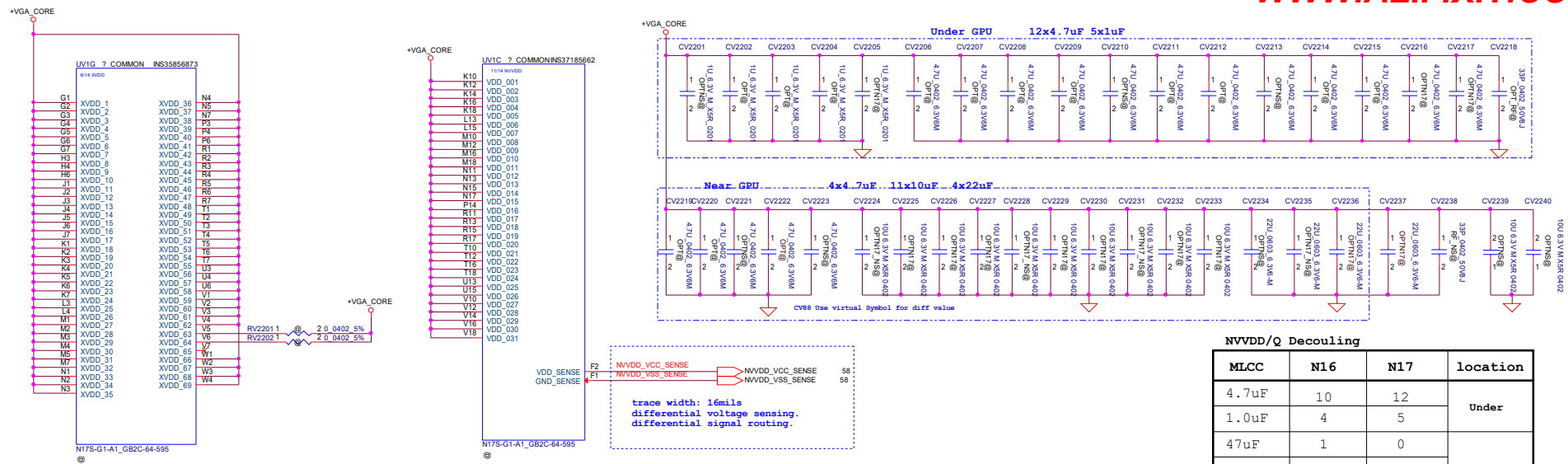
N16x Multi-level Straps

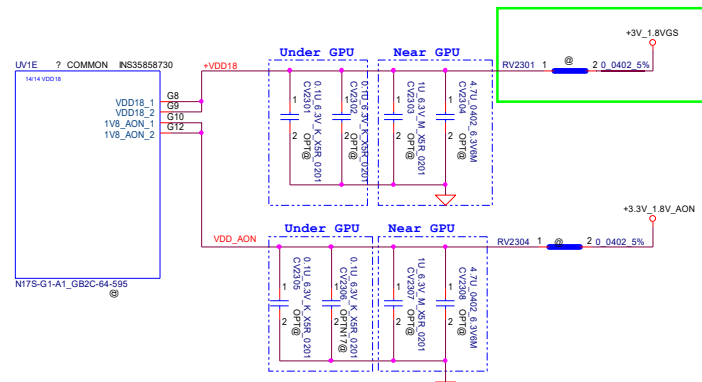
Physical Strapping pin	Power Rail	Logical Strapping Bit3	Logical Strapping Bit2	Logical Strapping Bit1	Logical Strapping Bit0
ROM_SCLK	+3VGS	SOR3_EXPOSED	SOR2_EXPOSED	SOR1_EXPOSED	SOR0_EXPOSED
ROM_SI	+3VGS	RAM_CFG[3]	RAM_CFG[2]	RAM_CFG[1]	RAM_CFG[0]
ROM_SO	+3VGS	DEVID_SEL	PCIE_CFG	SMB_ALT_ADDR	VGA_DEVICE
STRAP0	+3VGS	Reserved(keep pull-up and pull-down footprint and stuff 50Kohm pull-up)			
STRAP1	+3VGS				
STRAP2	+3VGS				
STRAP3	+3VGS				
STRAP4	+3VGS	Reserved(keep pull-up and pull-down footprint and not stuff by default)			





Title		
GPU_MEM Interface		
Size	Document Number	Rev
C	GS44D/GS54D	0.1





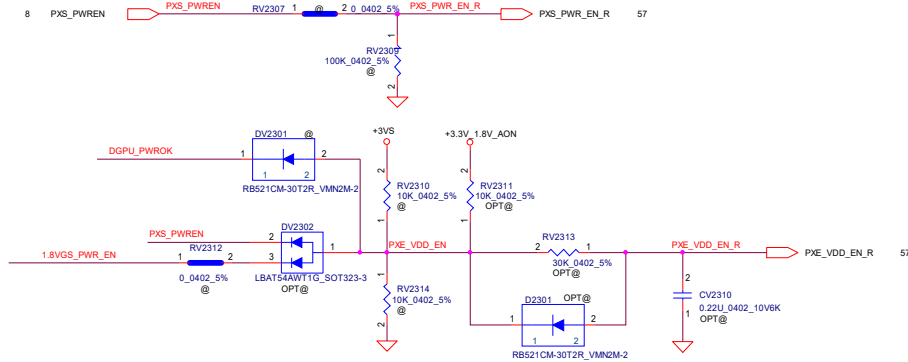
N16 3V3_MAIN(N17 VDD_18) Decoupling

MLCC	N16	N17	location
0.1uF	2	2	Under
1.0uF	1	1	Near
4.7uF	1	1	Near

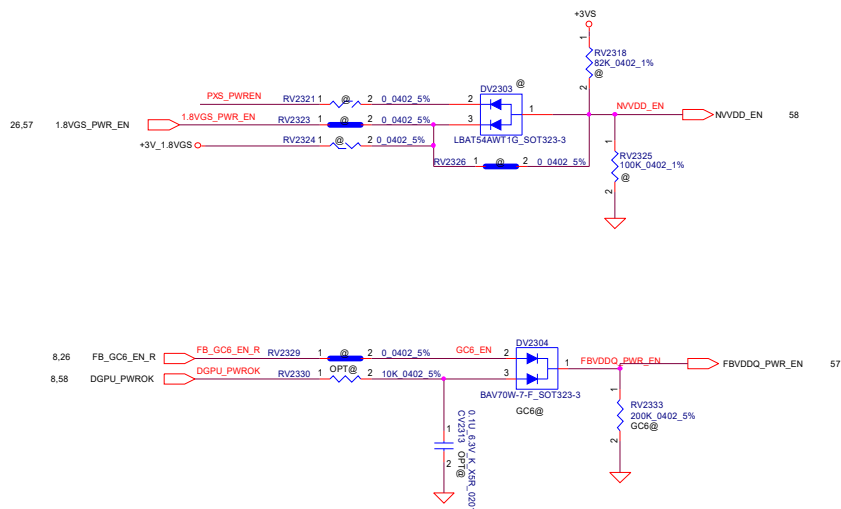
N16 3V3_AON(N17 1V8_AON) Decoupling

MLCC	N16	N17	location
0.1uF	1	2	Under
1.0uF	1	1	Near
4.7uF	1	1	Near

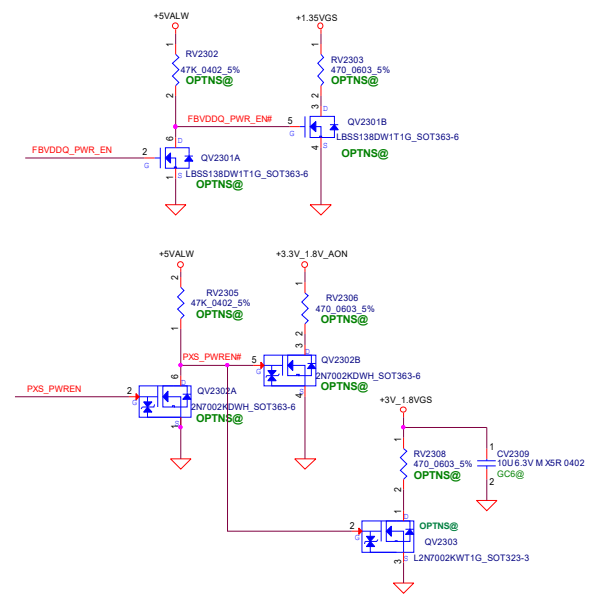
PXE_VDD & 1V8_AON



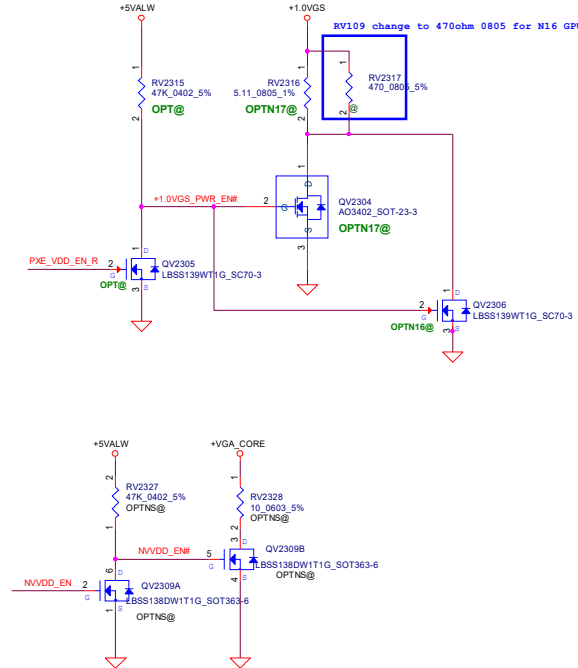
+1.8VG_AON TO +1.8VGS

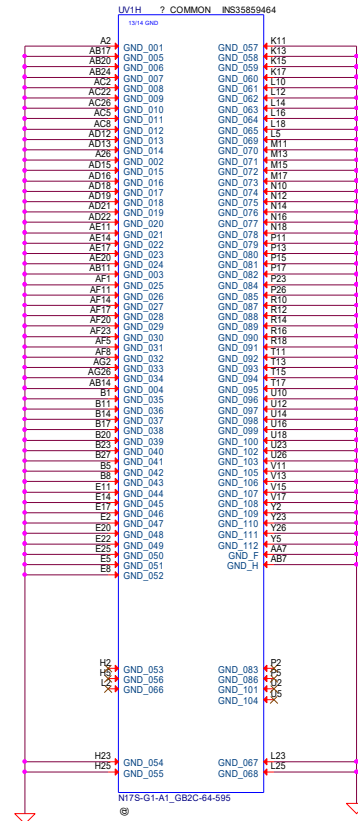


Discharge



RV2316 建虚拟料号, N16=470 ohm, N17=5.11 ohm



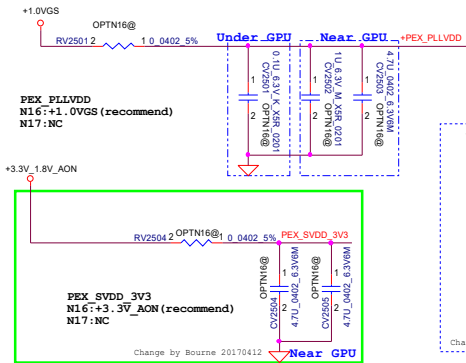


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					GS44D/GS54D
				Date	Friday, December 06, 2015
				Sheet	24 of 60
				Rev	0.1



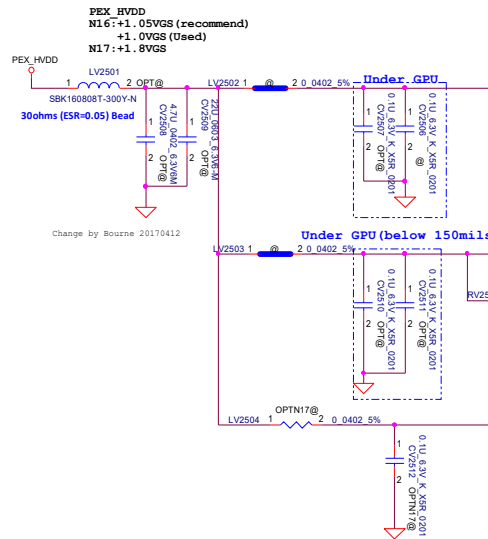
PEX_PLLVDD/Q Decoupling

MLCC	N16	N17	location
1.0uF	1	NA	Under
1uF	1	NA	Near
4.7uF	1	NA	



PEX_SVDD/Q Decoupling

MLCC	N16	N17	location
4.7uF	2	NA	Near



XS_PLLVDD/Q Decoupling

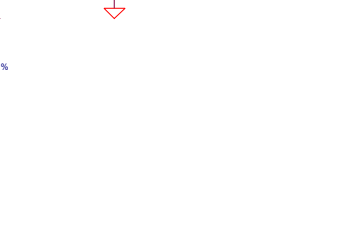
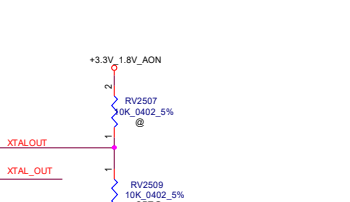
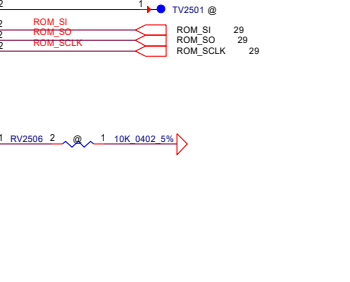
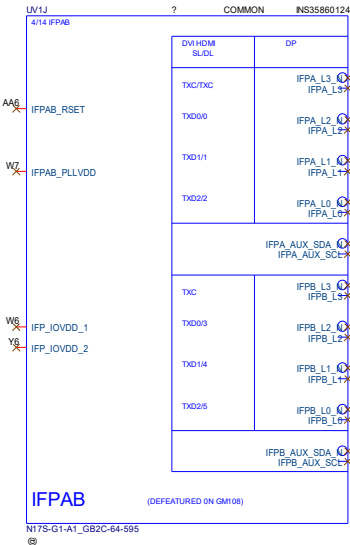
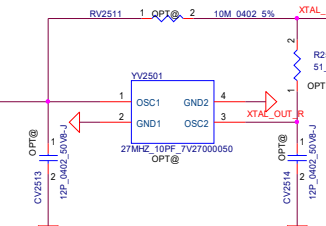
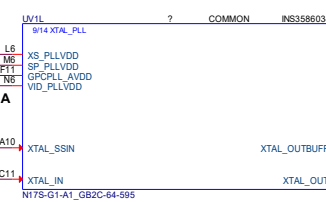
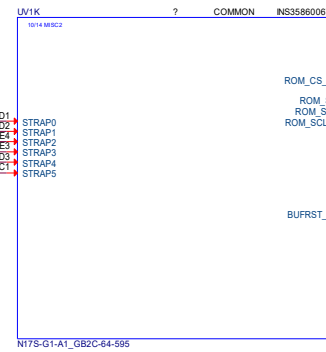
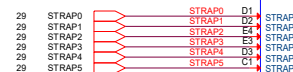
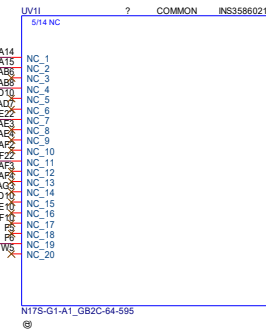
MLCC	N16	N17	location
0.1uF	1	1	Under
22uF	1	0	Near

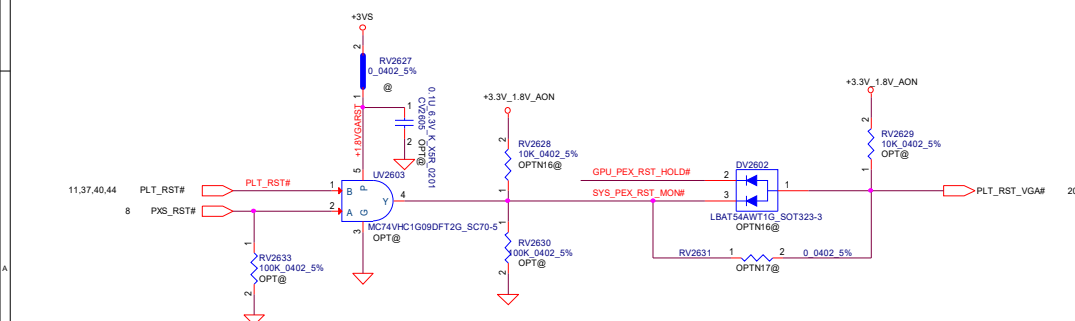
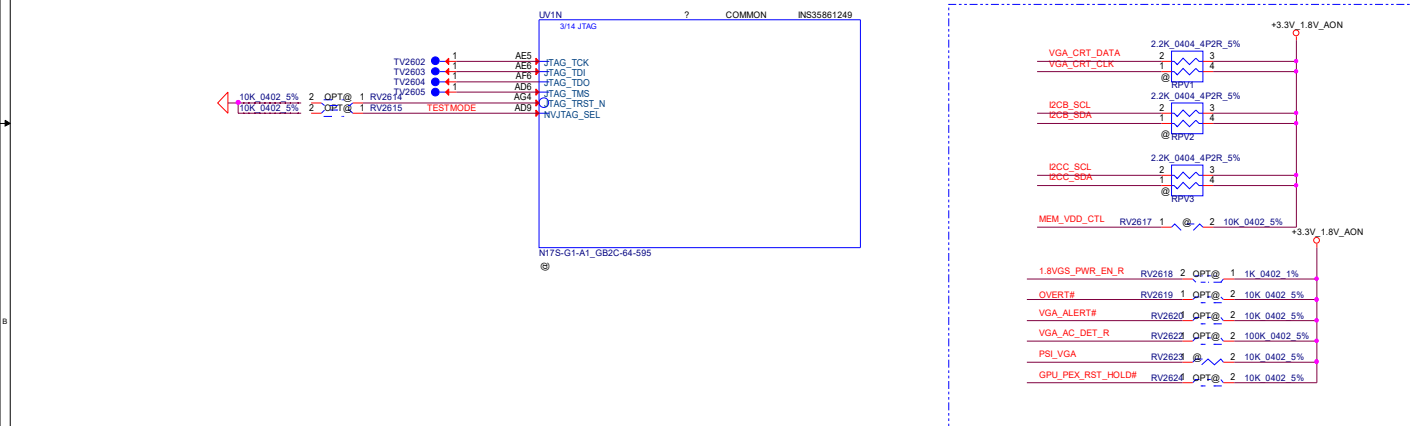
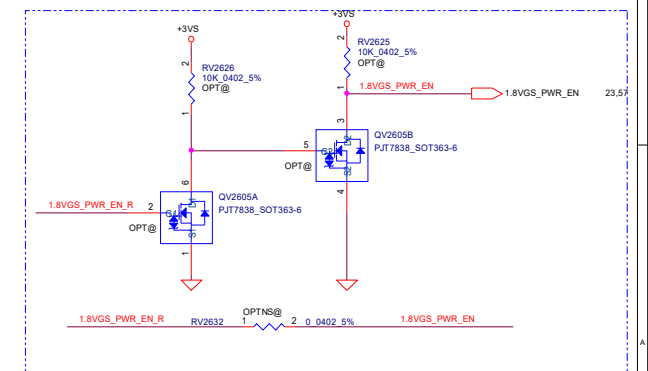
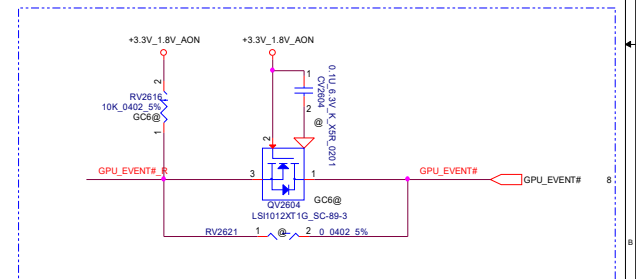
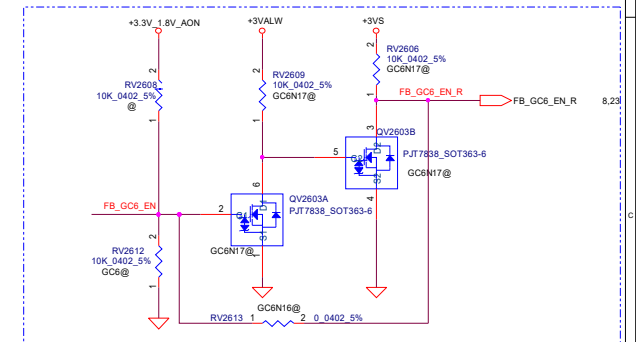
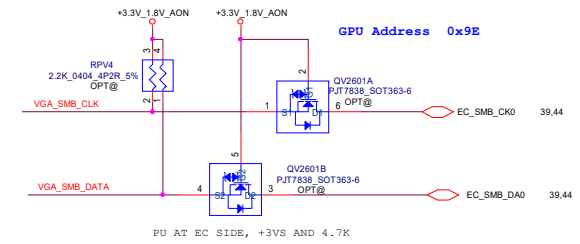
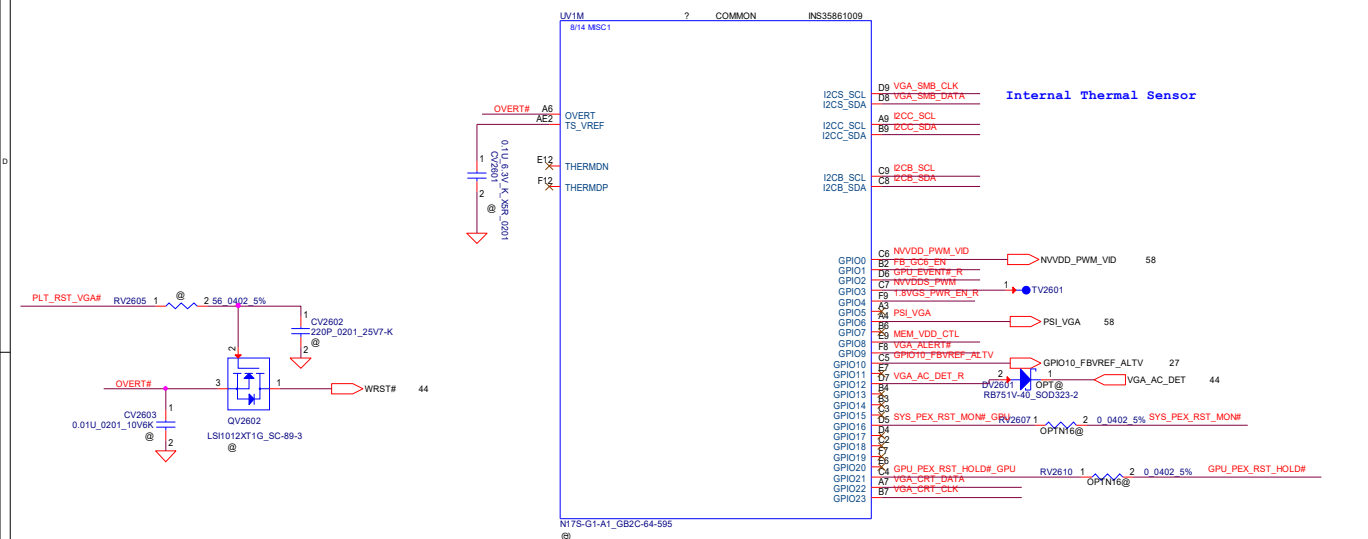
SP_PLLVDD & VID_PLLVDD/Q Decoupling

MLCC	N16	N17	location
0.1uF	2	2	Under
10uF	1	0	Near
47uF	1	0	

GPCPLL_AVDD/Q Decoupling

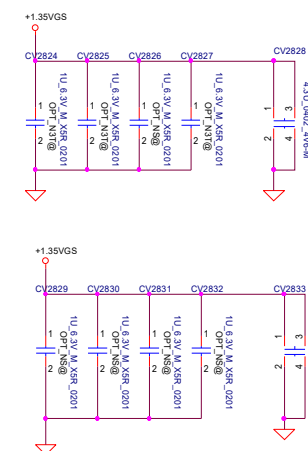
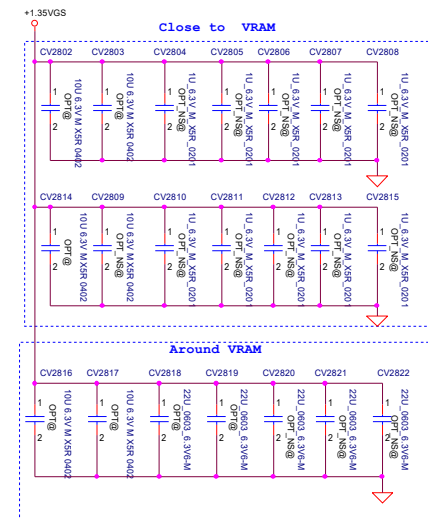
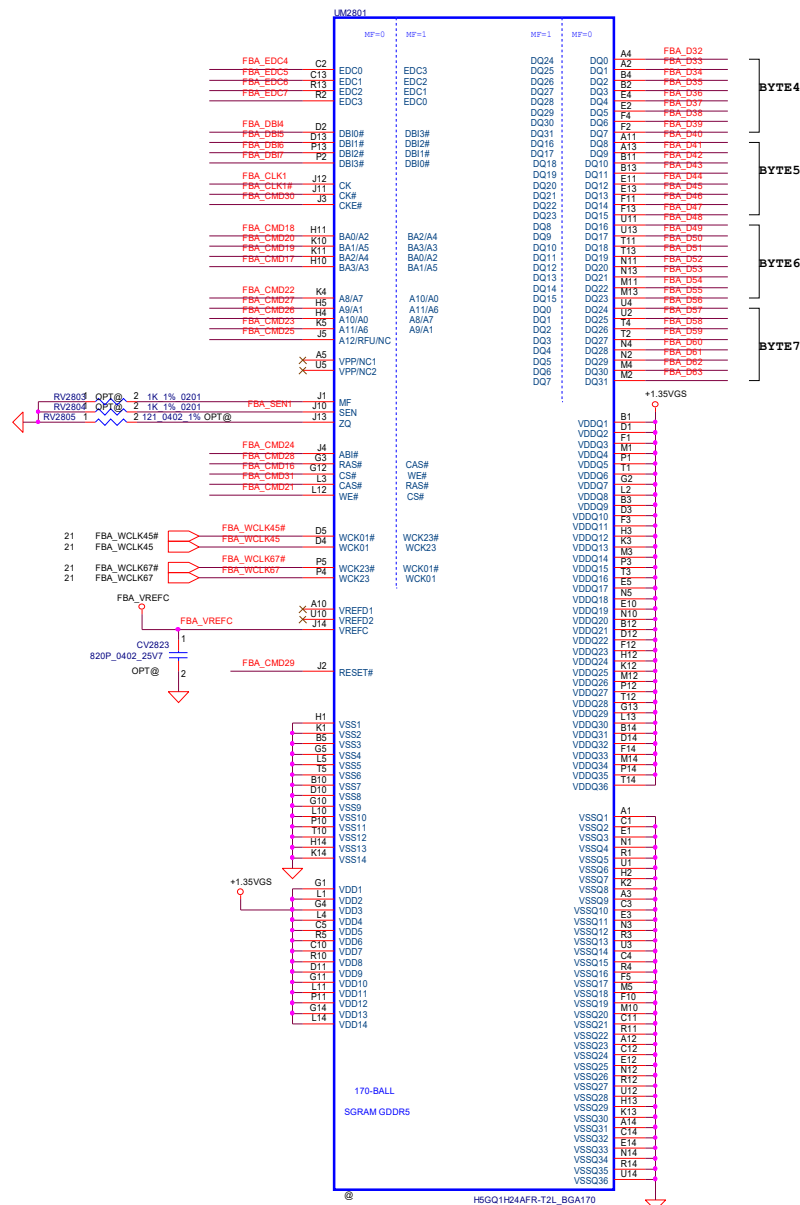
MLCC	N16	N17	location
0.1uF	NA	1	Under
4.7uF	NA	1	Near
22uF	NA	1	

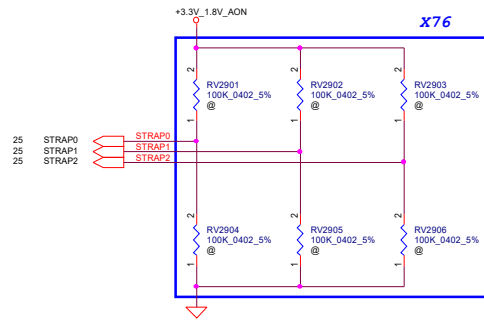




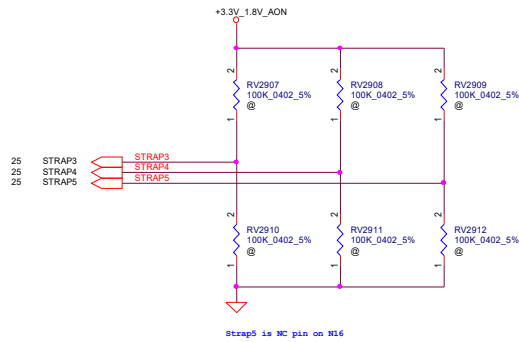
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Issued Date	2015/08/20	Deciphered Date	2019/08/20	GPU_GPIO/JTAG Size Document Number GS44D/GS54D	
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MF=0 No Mirror





GPU	FB Memory (GDDR5)	RAMCFG[4:0]	STRAP2	STRAP1	STRAP0
8Gb	Samsung 8Gb	K4G80325FB-HC28	0 (0x0000)	L	L
	Micron 8Gb	MT51J256M32HF-70:A	1 (0x0001)	L	L
	Hynix 8Gb	H5GC8H24MJR-R0C	2 (0x0010)	L	H



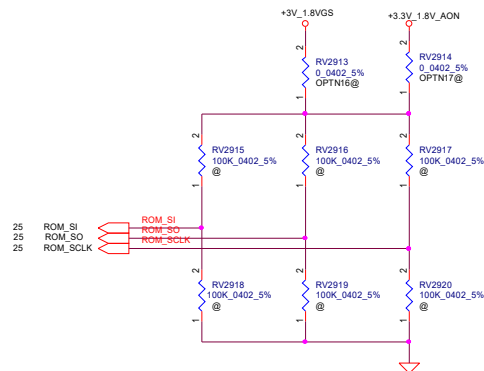
STRAP5	STRAP4	STRAP3	SMB_ALT_ADDR	DEVID_SEL	PCIE_CFG	VGA_DEVICE
L	L	L	0	0	0	0

1: SMB_ALT_ADDR ENABLE
0: SMB_ALT_ADDR DISABLE

1: DEVID_SEL REBRAND
0: DEVID_SEL ORIGINAL

1: PCIE_CFG LOW POWER
0: PCIE_CFG HIGH POWER

1: VGA_DEVICE ENABLE
0: VGA_DEVICE DISABLE



	ROM_SO	ROM_SI	ROM_SCLK	SOR_EXPOSED[3:0]
N17S-G1	H	H	M	0000
N16S-GTR				

1: ENABLE 0: DISABLE

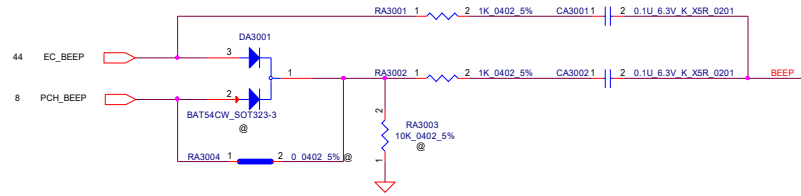
SOR0/1/2/3 DISABLE

DEVID_SEL	
0	(Default)
1	

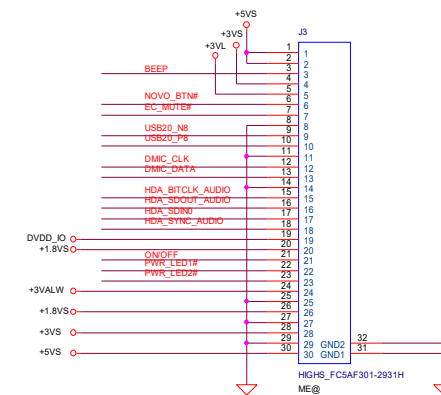
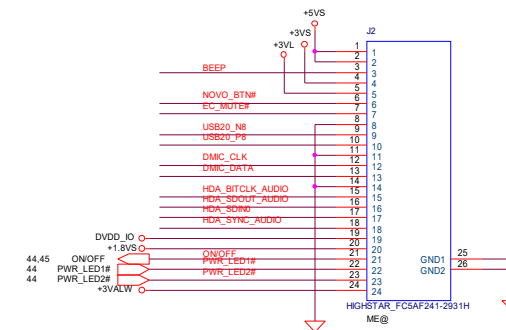
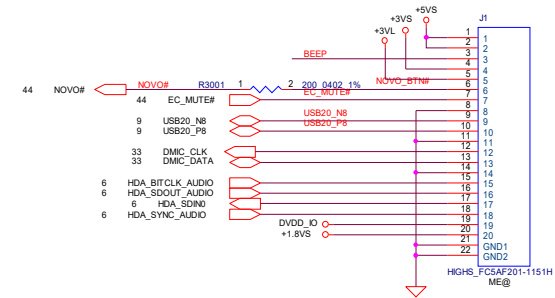
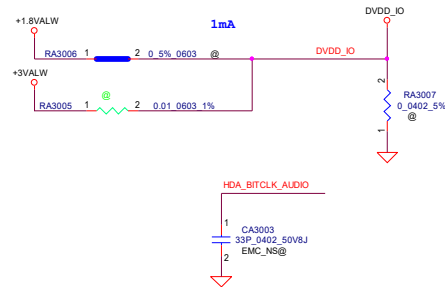
PCIE_CFG	
0	(Default)
1	

SMBUS_ALT_ADDR	
0	0x9E (Default)
1	0x9C (Multi-GPU usage)

VGA_DEVICE	
0	3D Device (Class Code 302h)
1	VGA Device (Default)



CPU HDA BUS power 1.8VALW



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				Date: Friday, December 06, 2016	Sheet 30 of 60



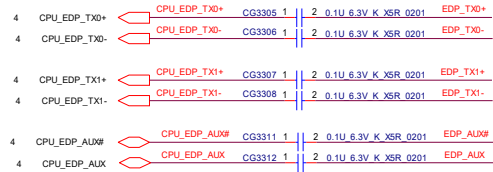
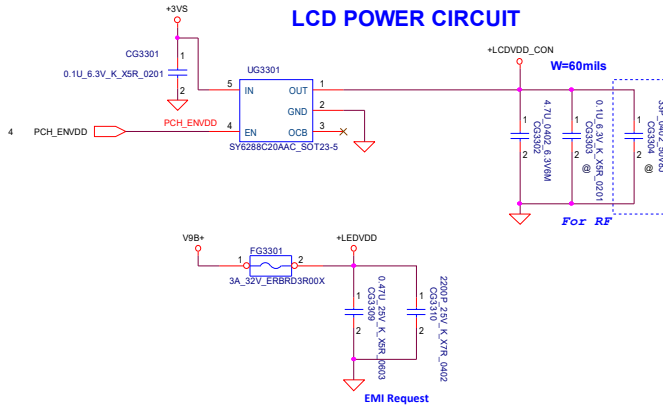
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				Date:	Friday, December 06, 2019
				Sheet	31 of 60
				Rev	0.1



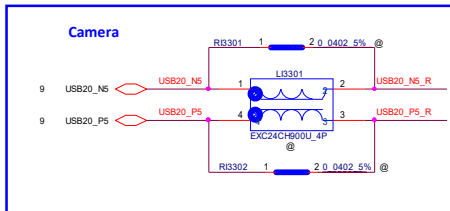
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Size	Document Number			Rev	
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Date:		Friday, December 06, 2019	Sheet	32	of 60



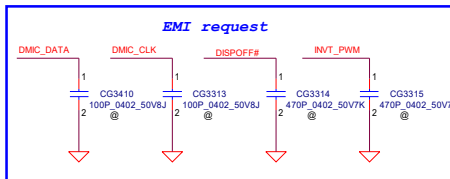
LCD POWER CIRCUIT



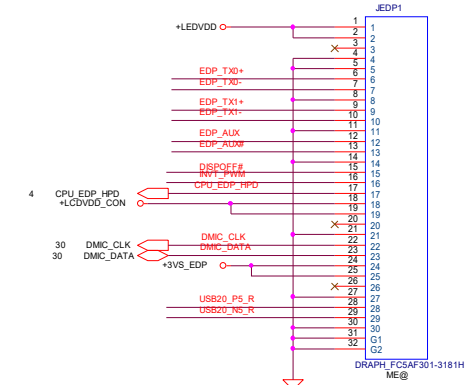
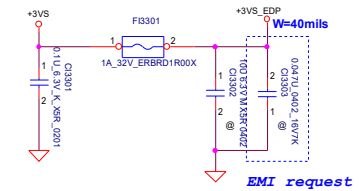
Camera



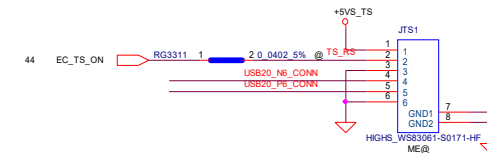
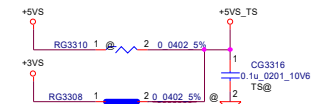
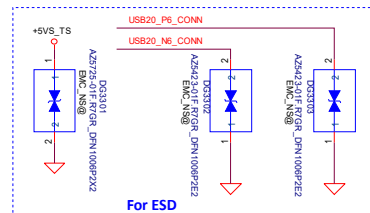
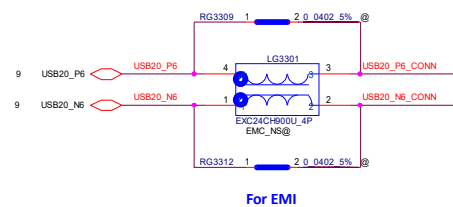
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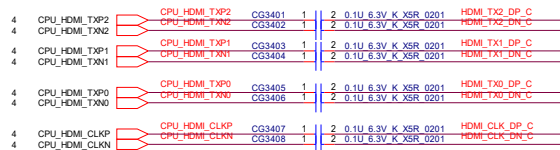


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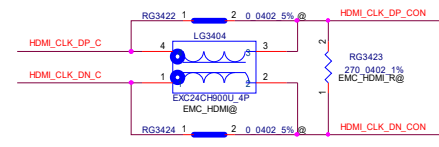
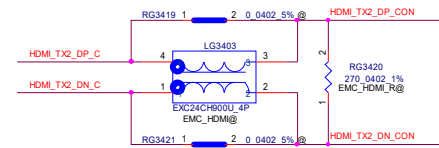
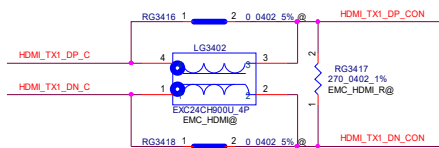
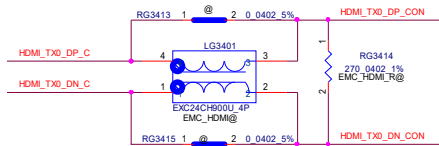
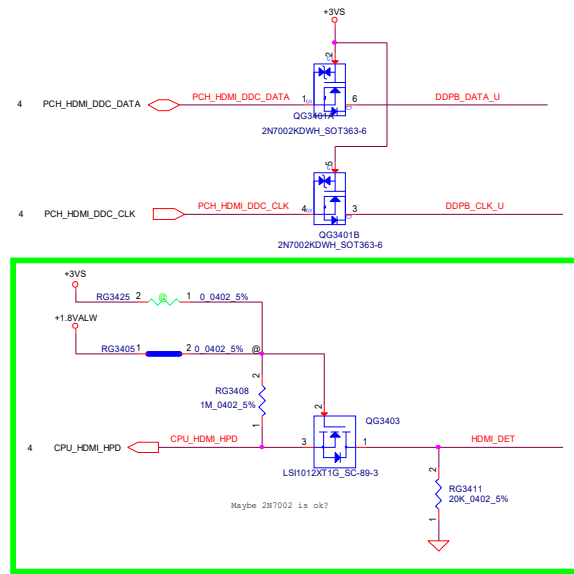
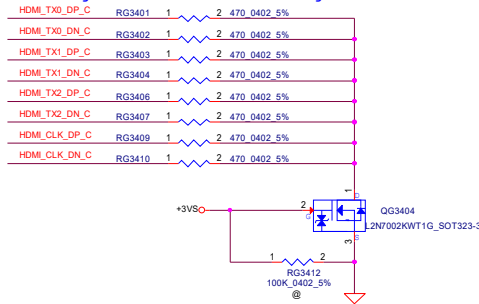


Touch Screen

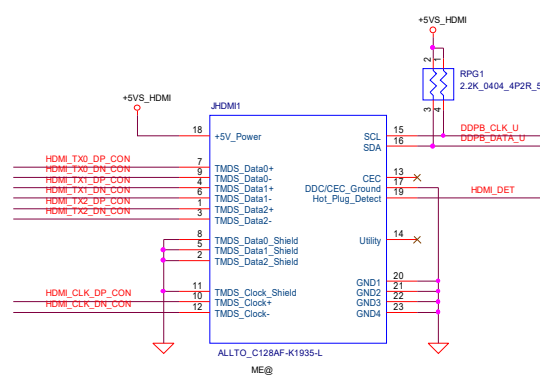
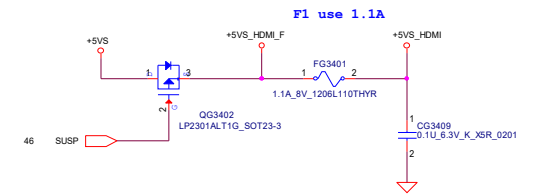
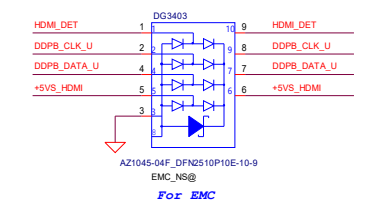
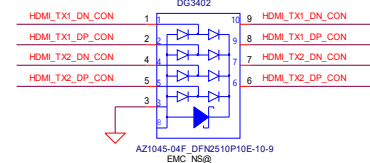
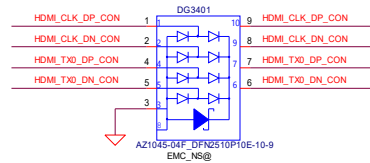




Need to change about 4700hm 5%-575412 Page115 Rev0.8



For EMC



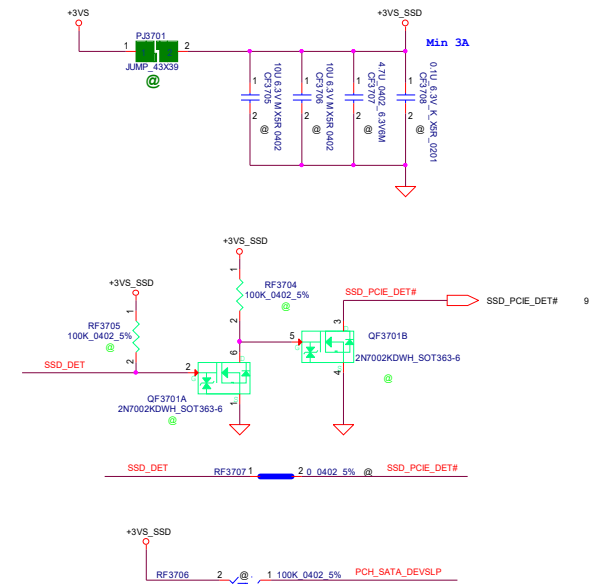
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Issued Date	2015/08/20	Deciphered Date	2016/08/20	Size	Document Number
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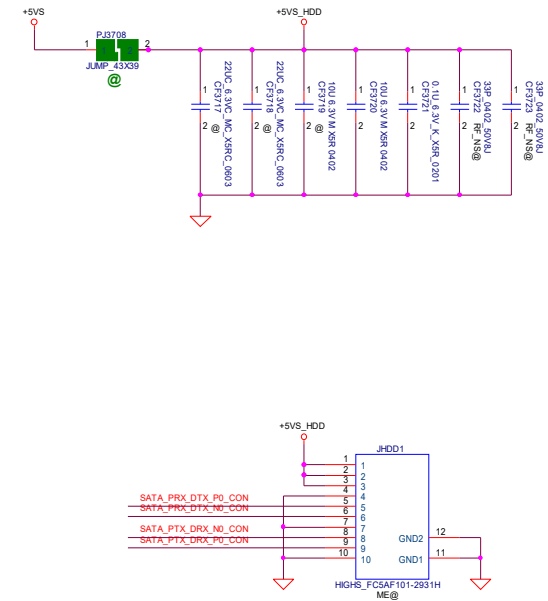


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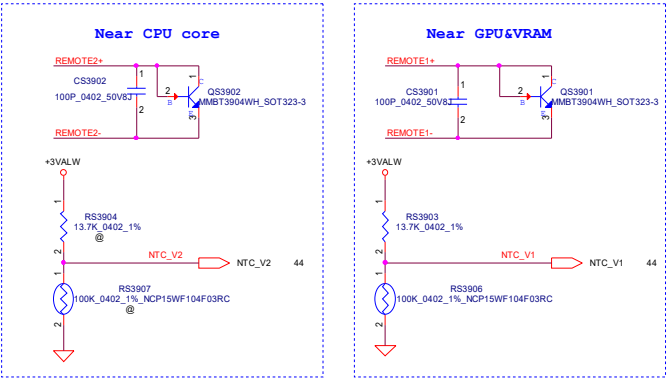
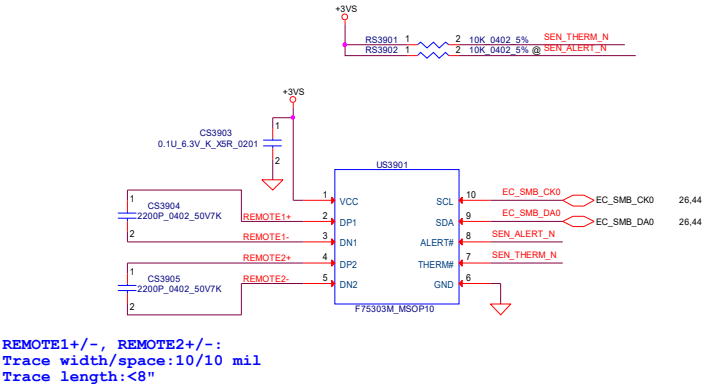


SATA Redriver

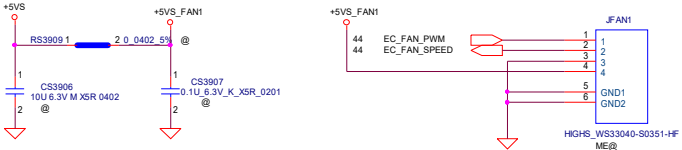


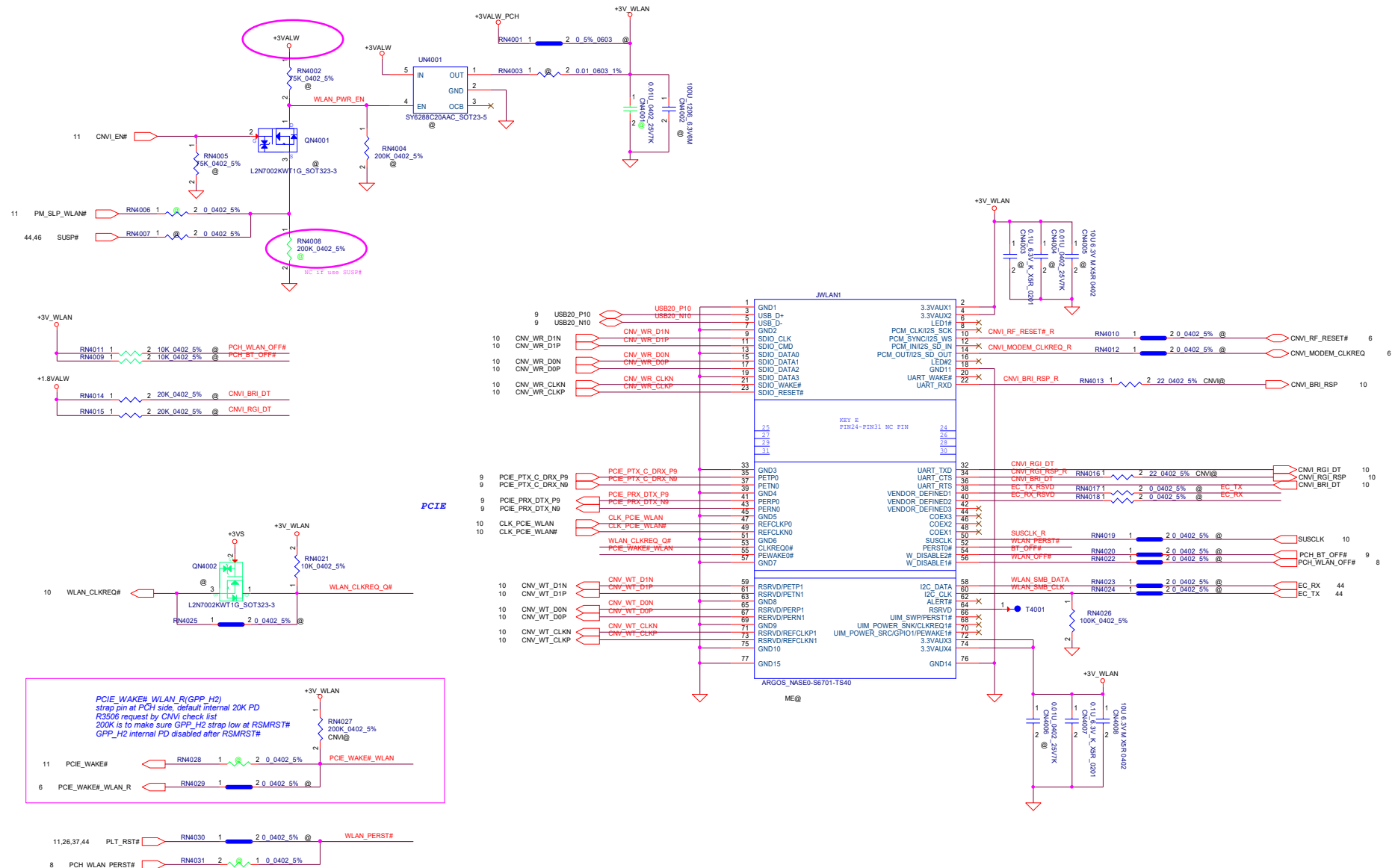
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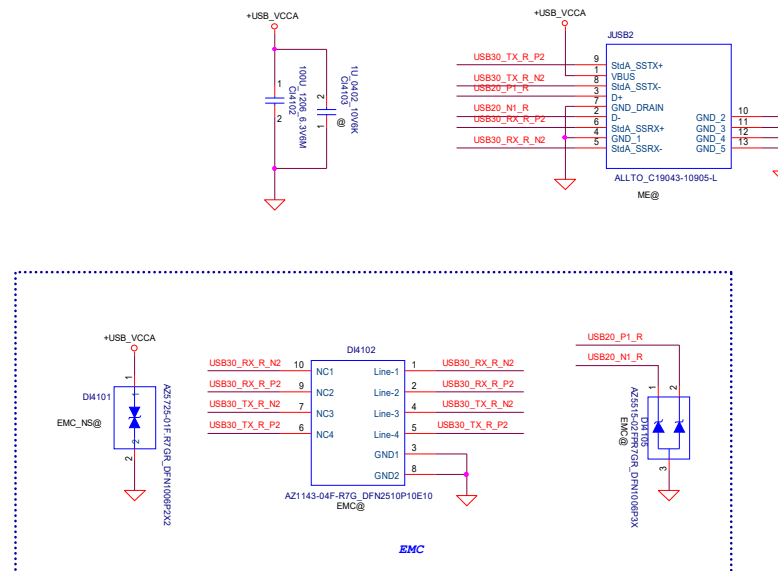
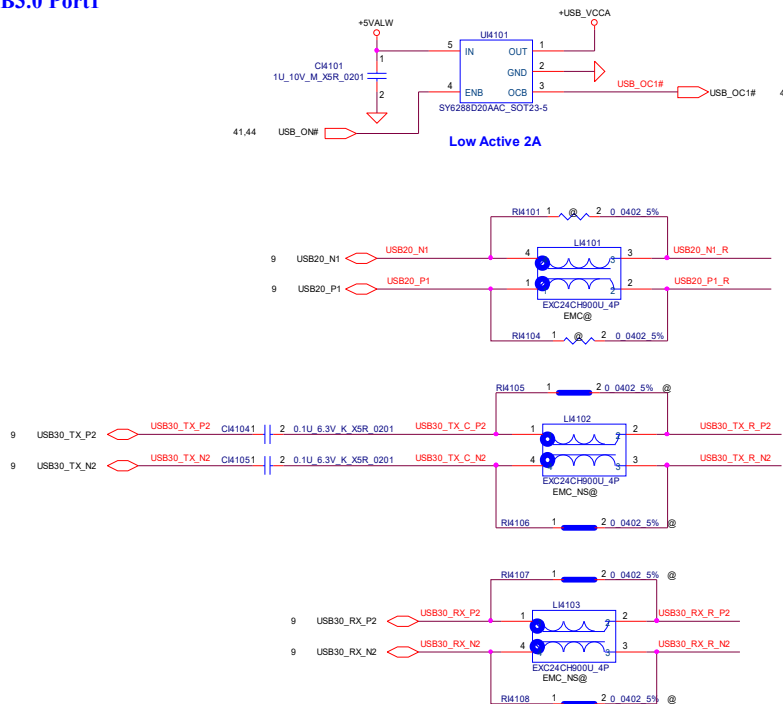


FAN Conn

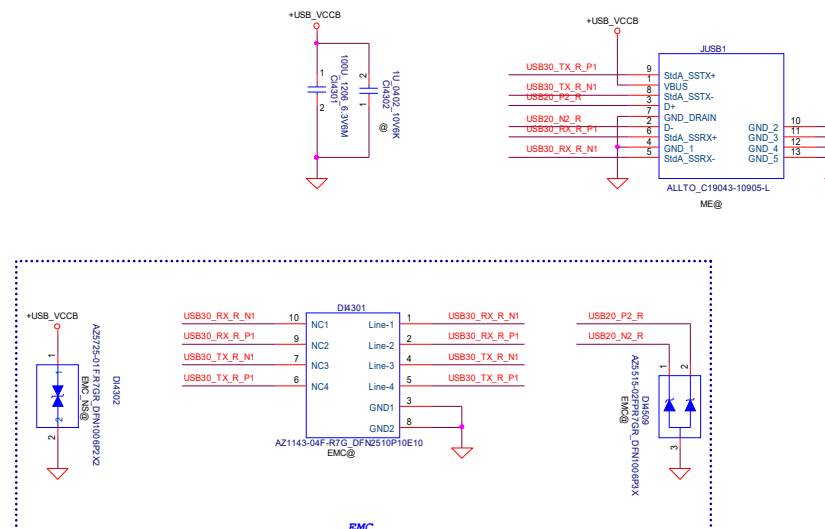
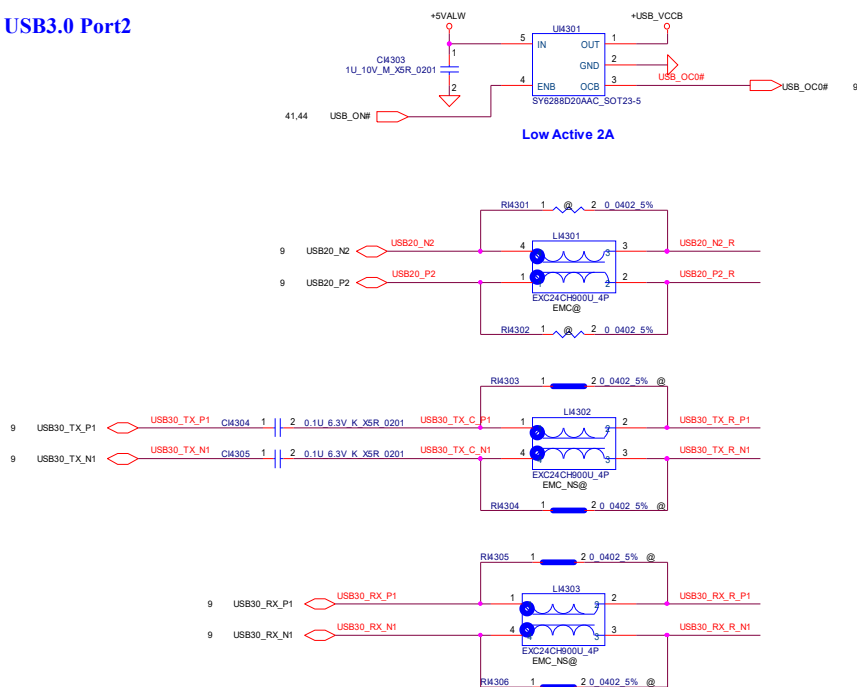




USB3.0 Port1



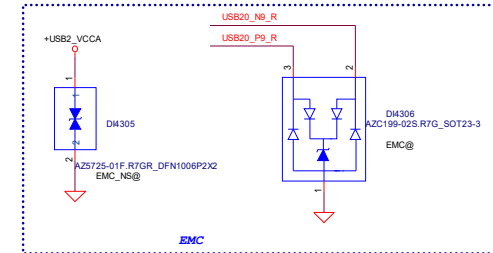
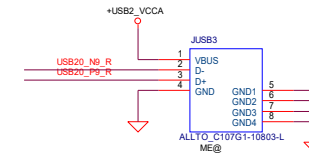
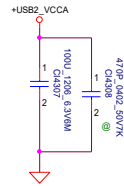
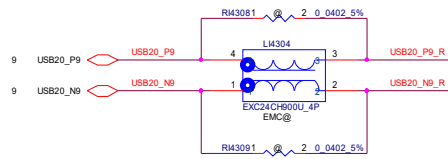
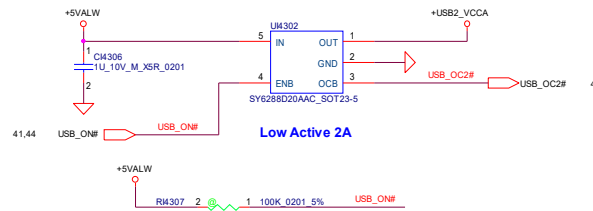
USB3.0 Port2



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USB2.0 PORT x1

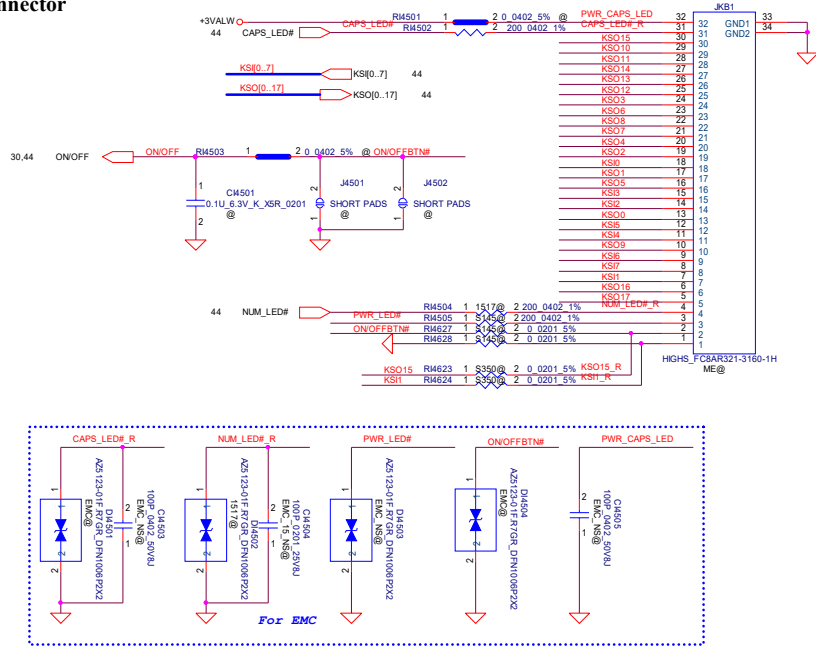


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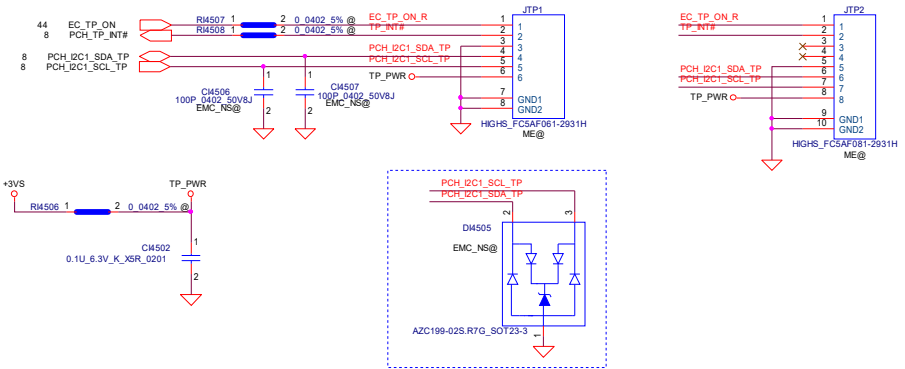


K/B Connector

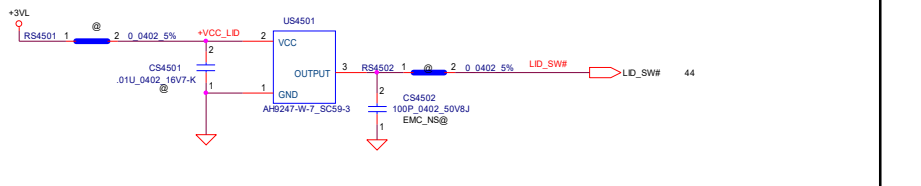


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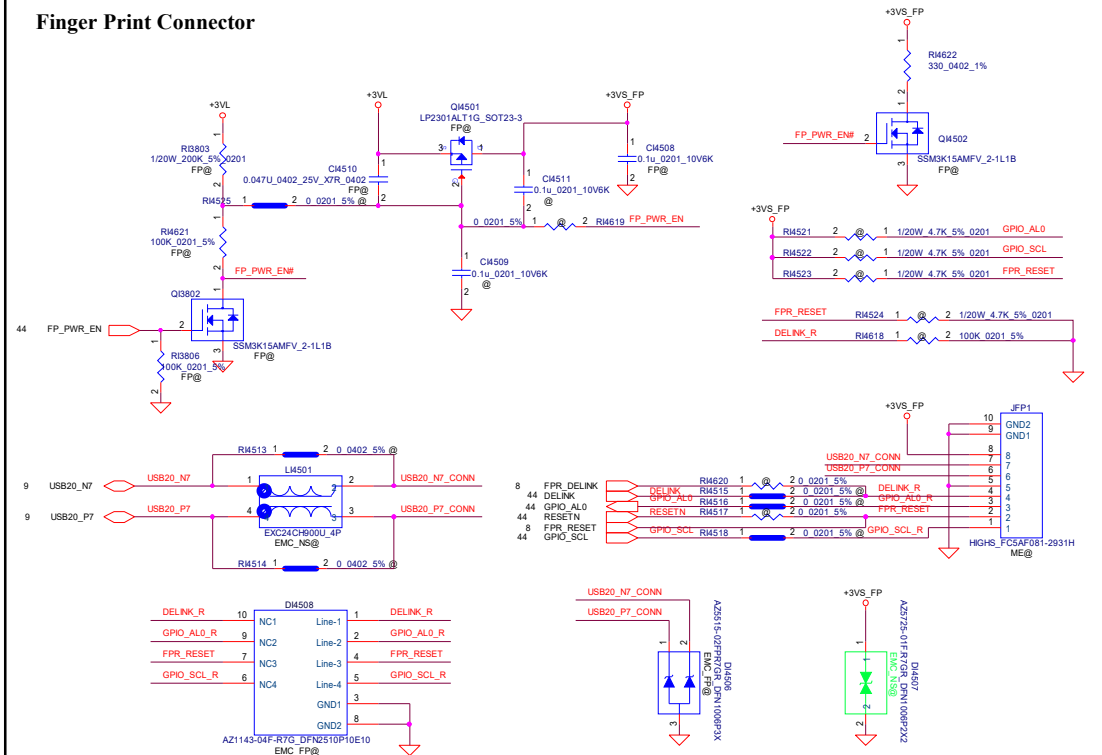
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HALL Sensor

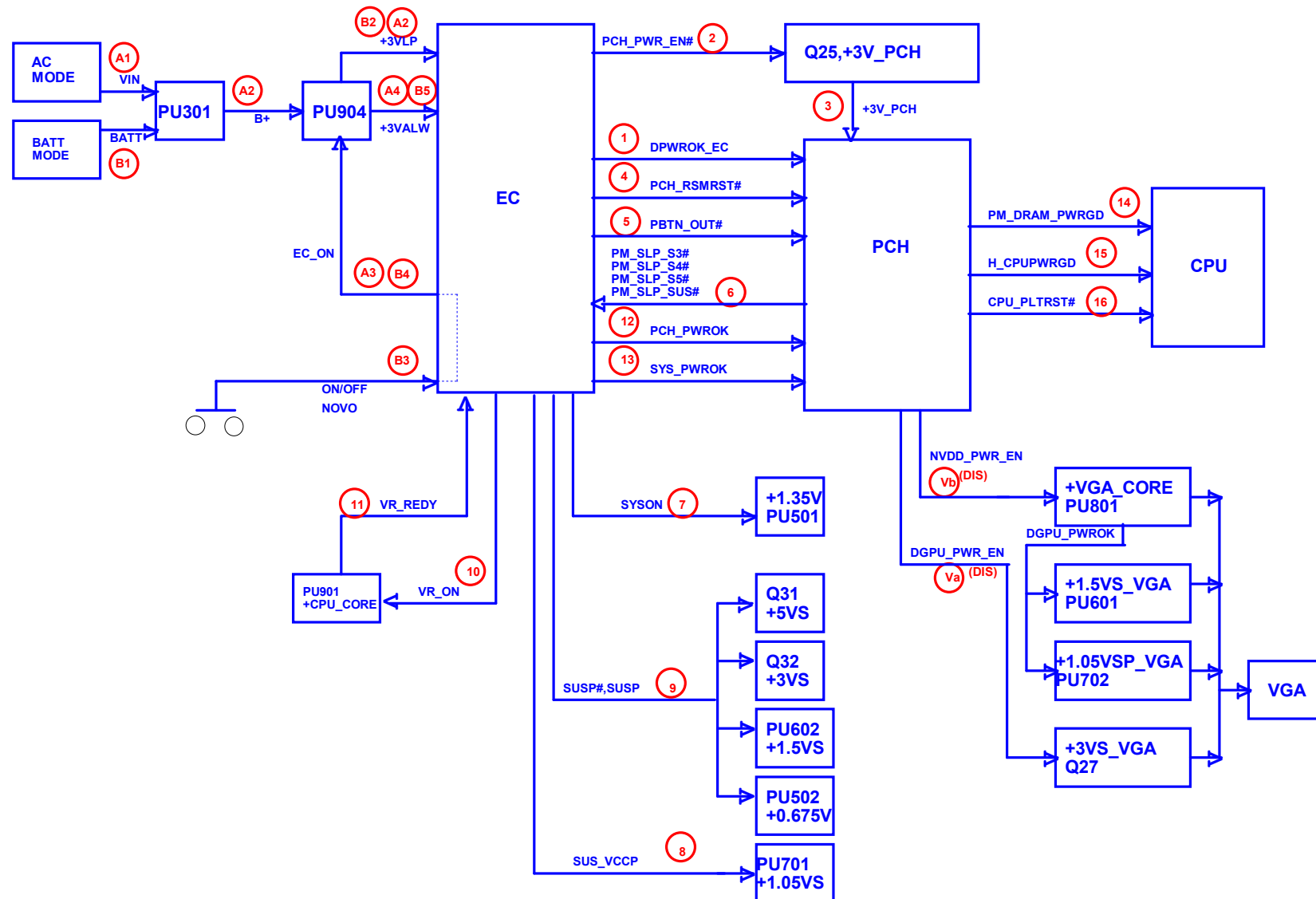


Finger Print Connector



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Power sequence block		0.1	
Size	Document Number	GS44D/GS54D	
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CPU



DRAM_Samsung



DRAM_Hynix

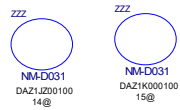


DRAM_Micron



SO_DIMM only

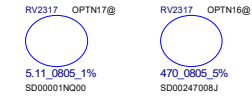
PCB_MB



HDMI_Royalty



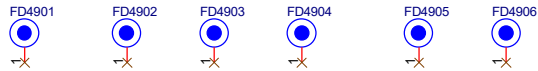
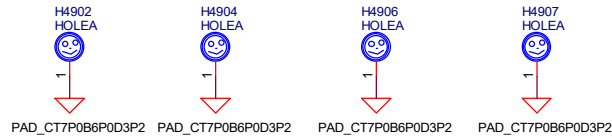
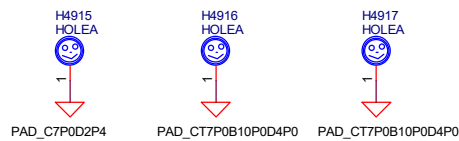
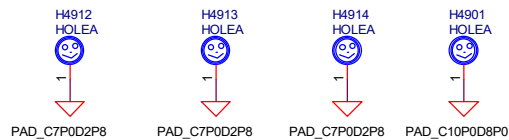
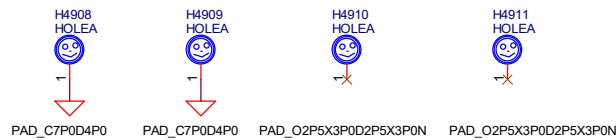
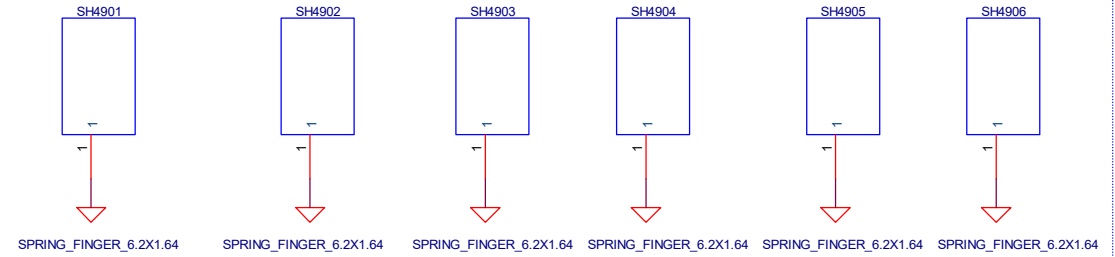
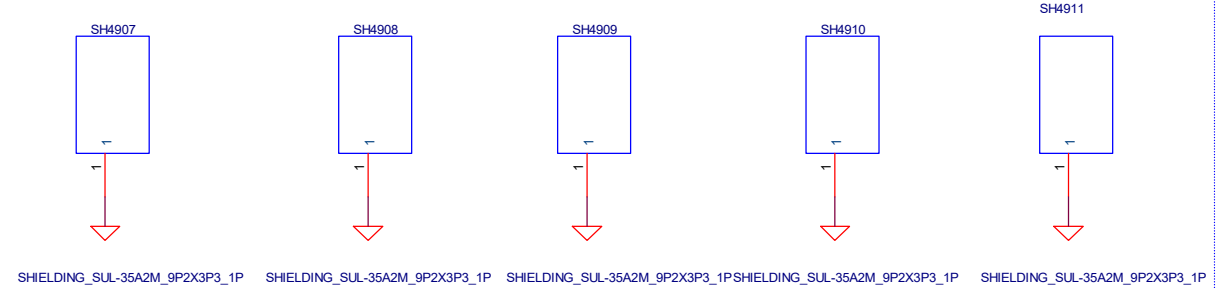
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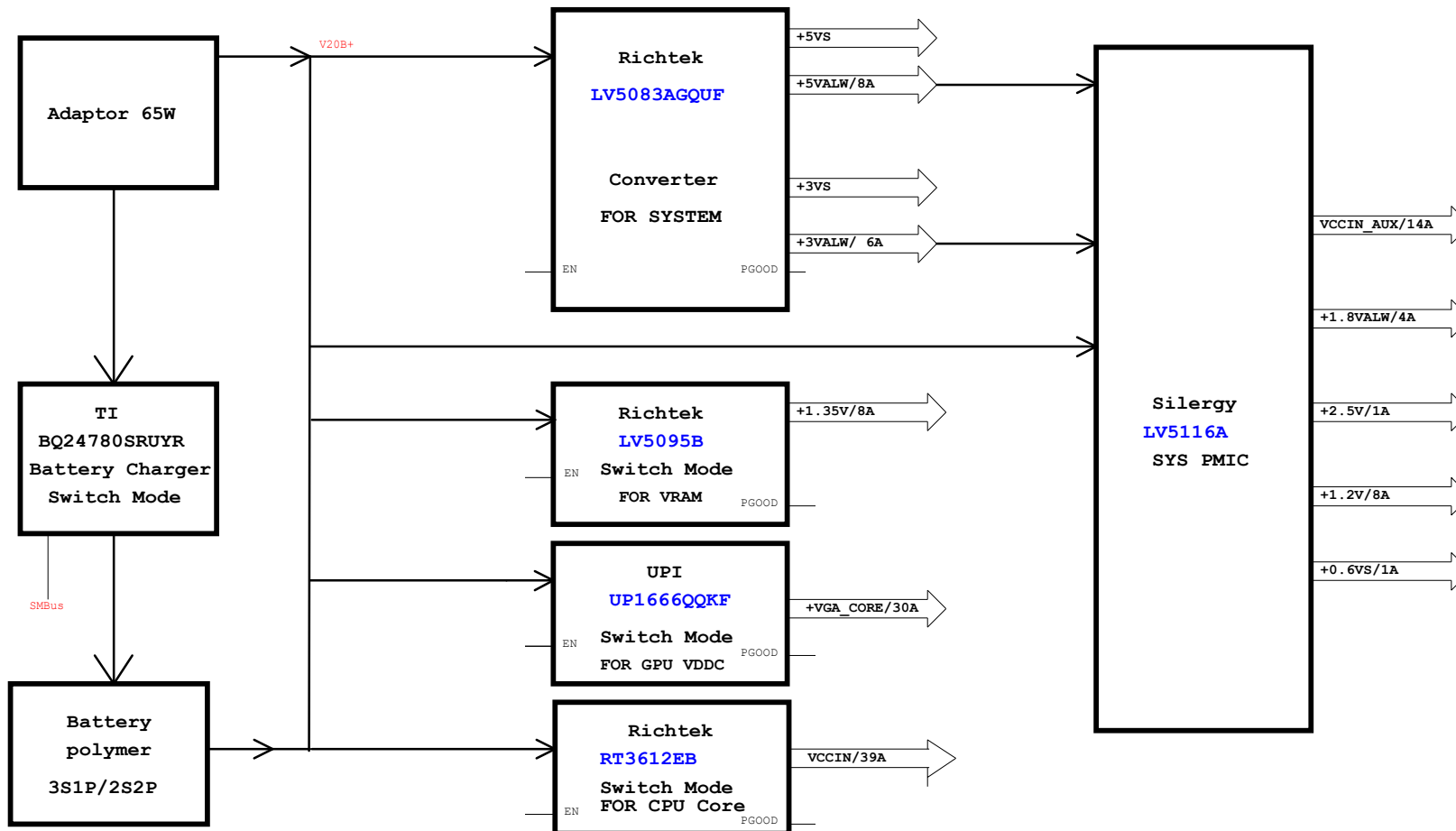
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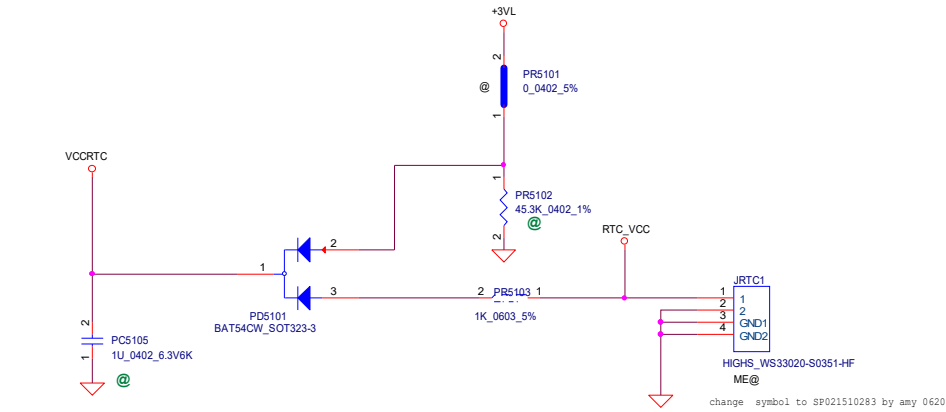
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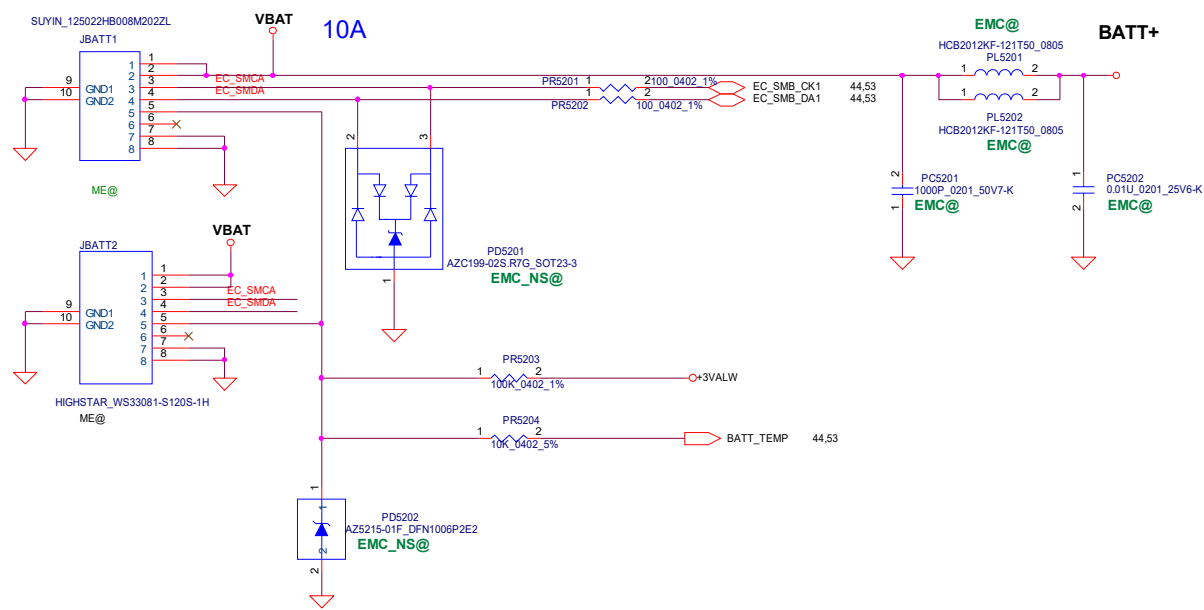
PCB Fedical Mark PAD**CPU Thermal Hole****GPU Thermal Hole****MD Shielding****SODIMM Shielding**

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				5	0.1	
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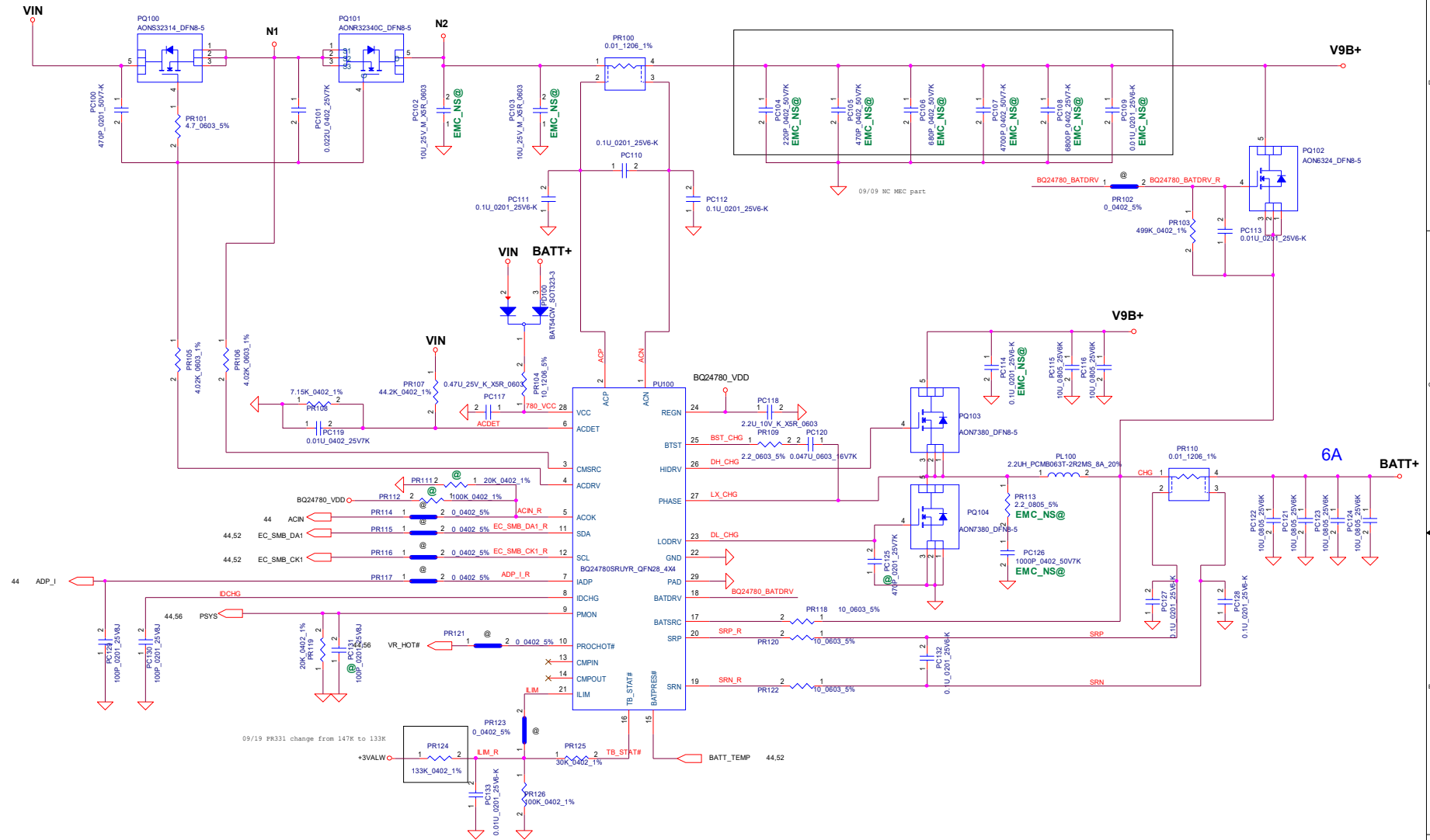
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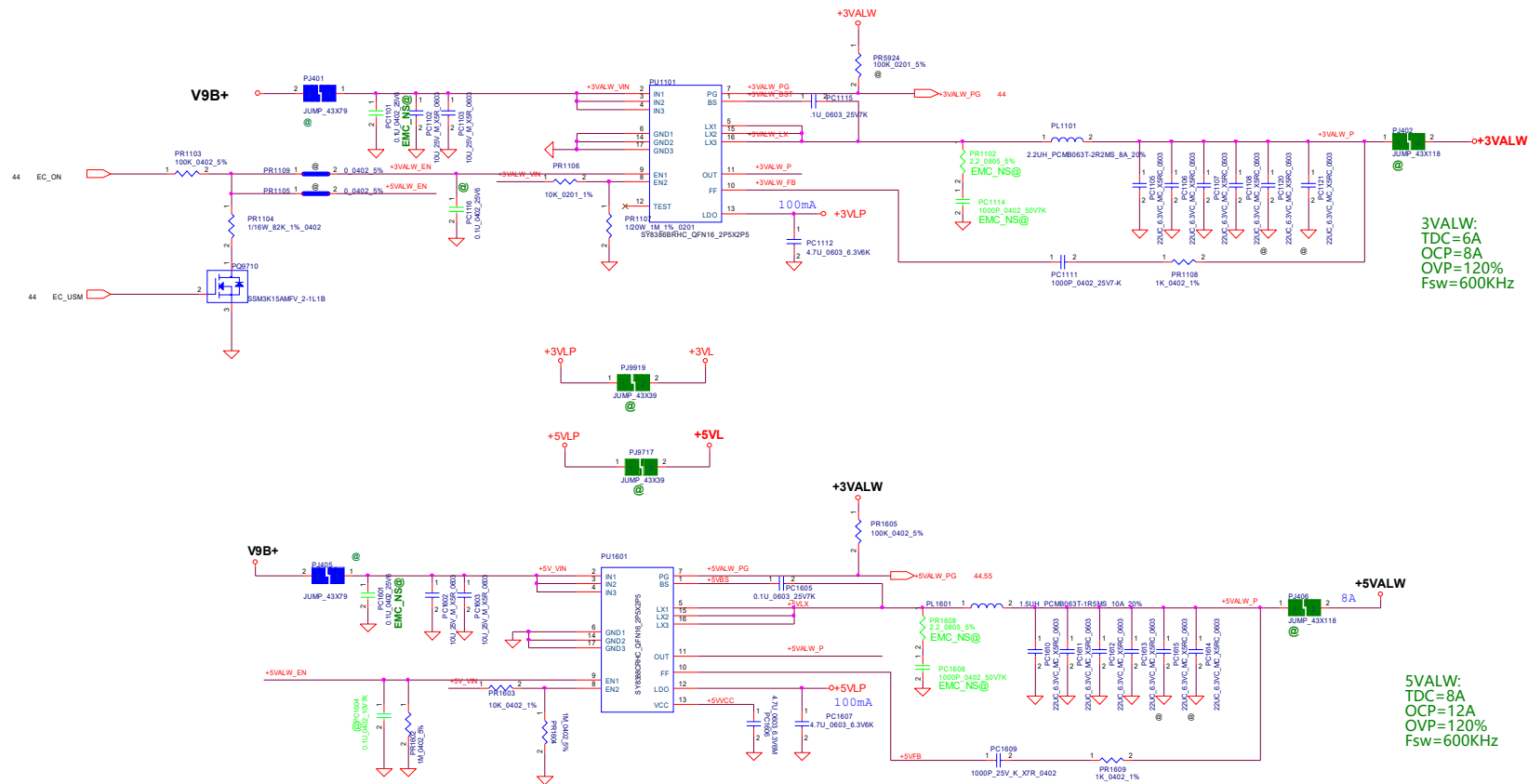


2S1P polymer battery
voltage level: +5.5V ~
8.8 V

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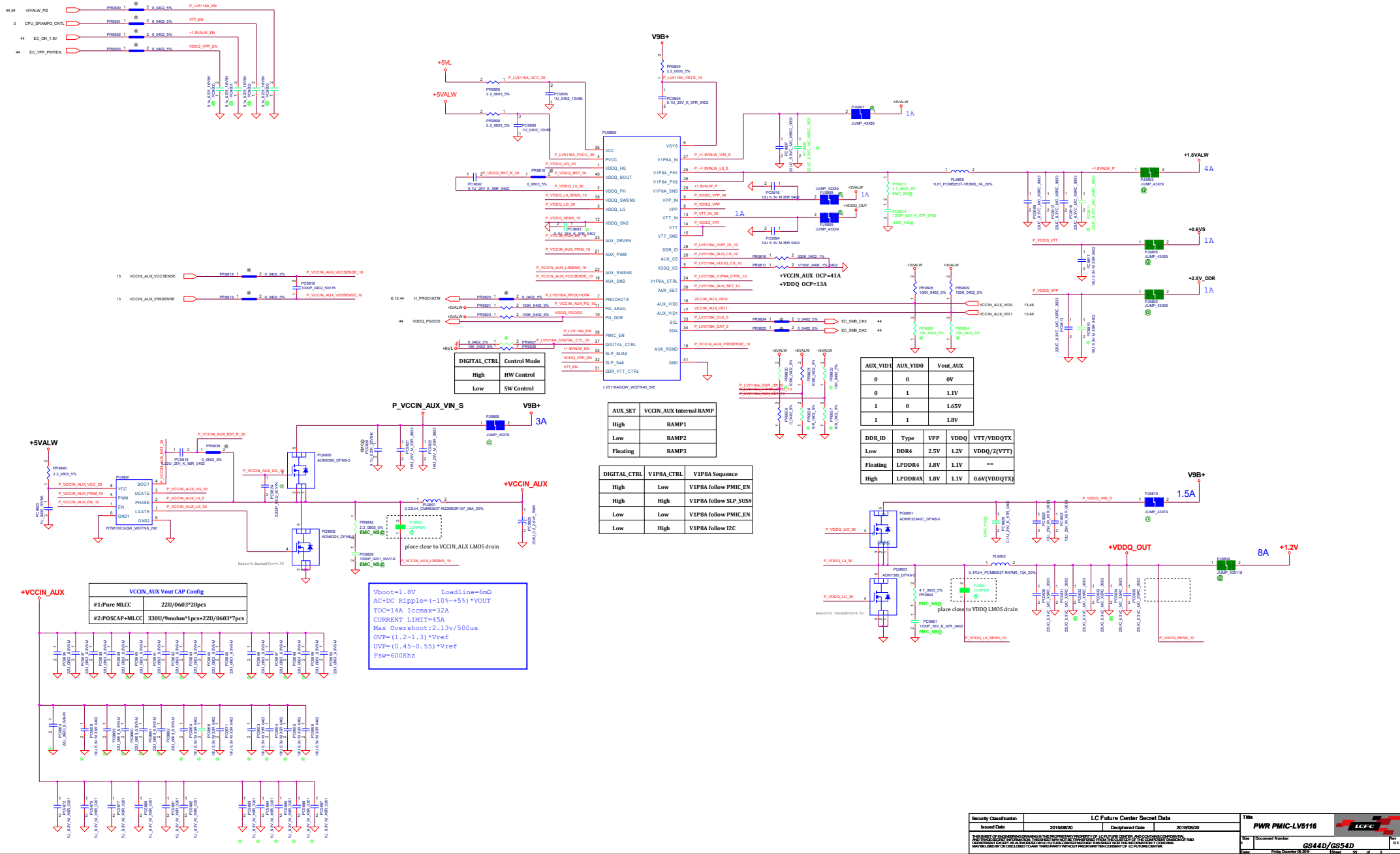
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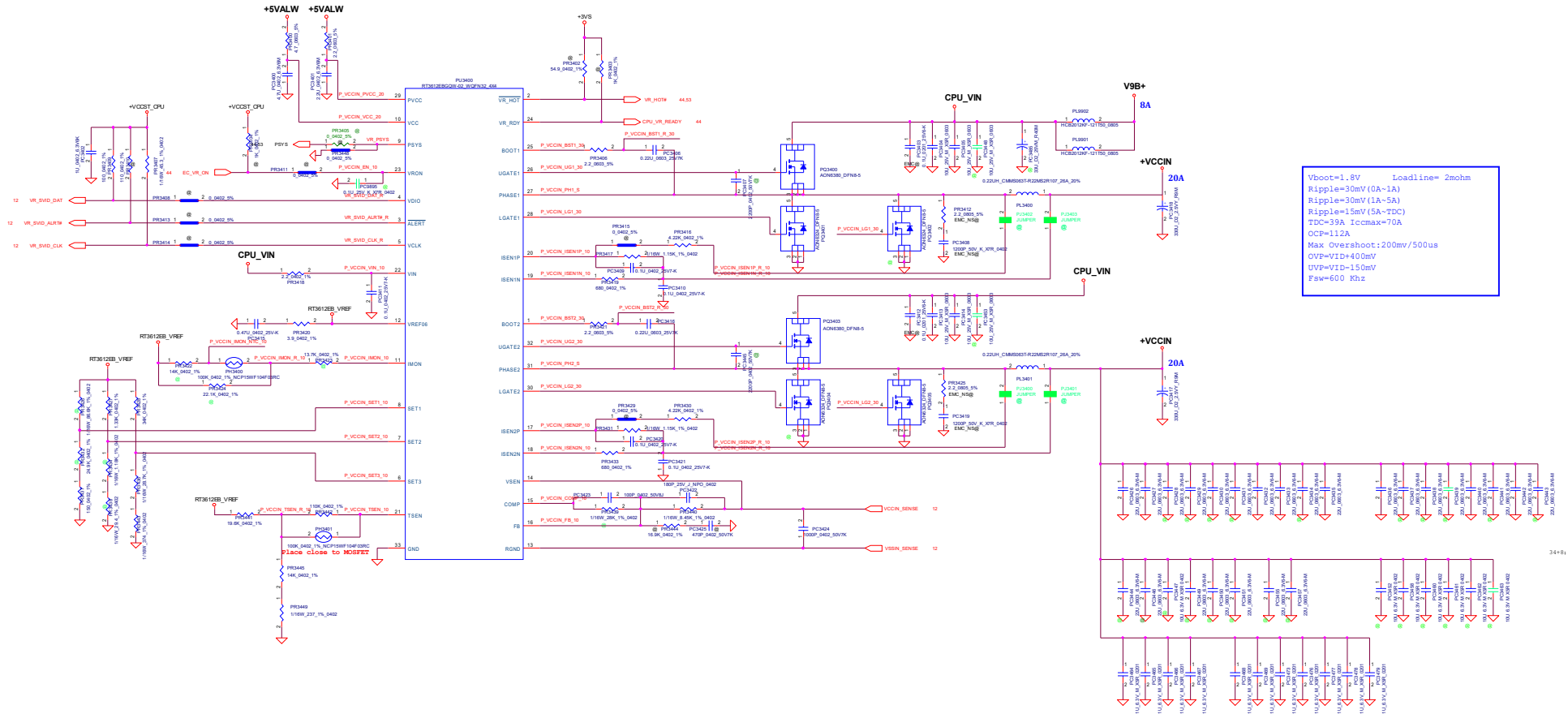


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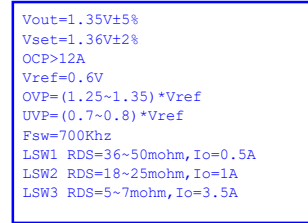




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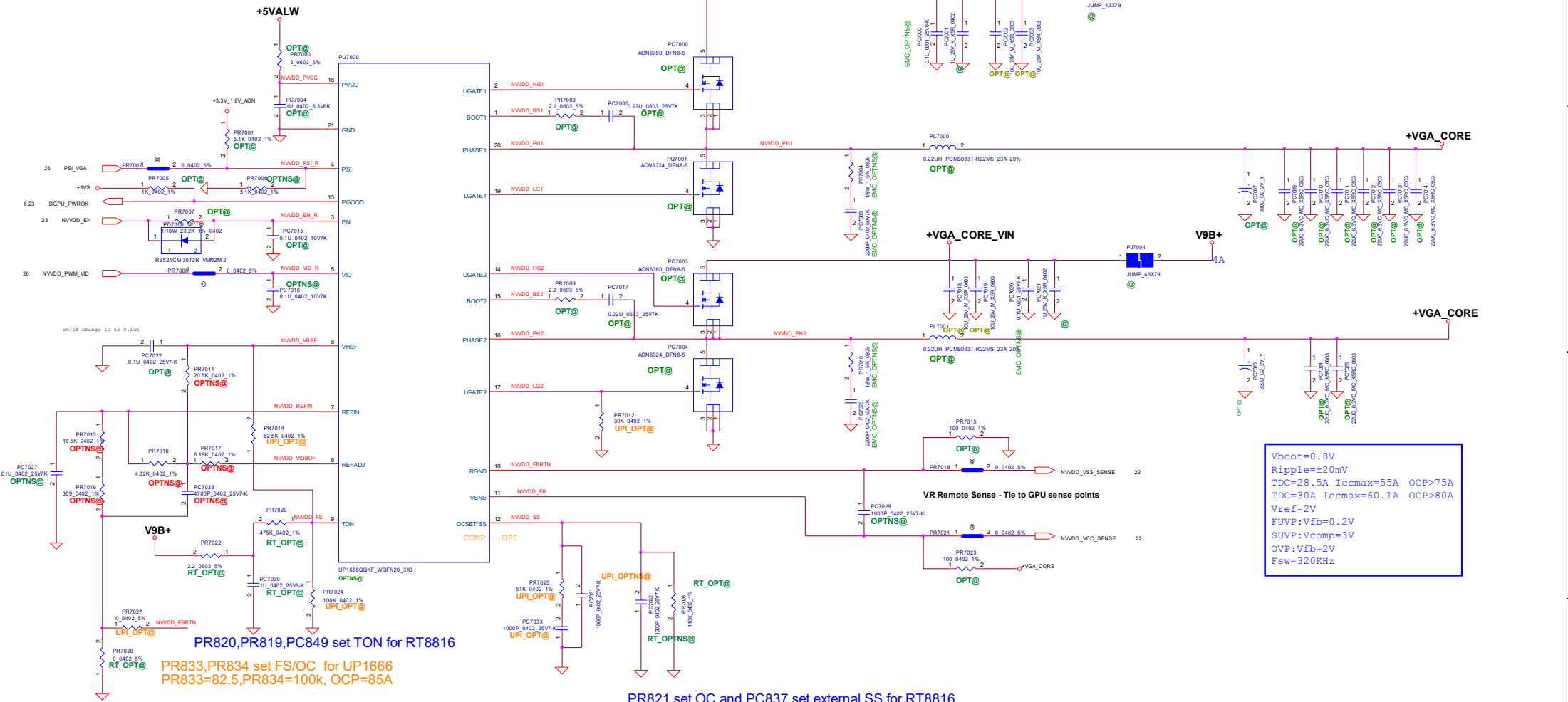


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Date		2019/08/29		Rev	
				0.1	



PWM-VID Specification			
	N17 Config	N16 Config	B
Vmin(V)	0.3	0.6	
Vmax(V)	1.3	1.2	
Vboot(V)	0.8	0.9	
Vstep(mV)	6.25	6.25	
N(level)	160	96	
Fpwm(KHz)	675	1.125	
Tdmin(nS)	9.26	9.26	
T(us)	<100	<100	

RT8816 PSI	UPL666 PSI	Phase Configuration
1.6V~5.5V	1.6~5.5V	2Phase CCM
1.08~1.35V	1~1.4V	2Phase DEM
0.7~0.88V	0.4V~0.8V	1Phase CCM
0~0.4V	0~0.2V	1Phase DEM



PR816,PR812,PR815,PR813,PR818,PC826 BOM structure control for N16 or N17

Component	Value	N17	N16
R1 (KΩ)	PR7017	6.19	20
R2 (KΩ)	PR7011	20.5	20
R3 (KΩ)	PR7016	4.32	2
R4 (KΩ)	PR7013	16.5	18
R5 (KΩ)	PR7019	0.309	0
C (nF)	PC7028	4.7	2.7

UPI_OPT@ : for UPL666
RT_OPT@ : for RT8816A

Vboot=0.8V
Ripple=±20mV
TDC=28.5A Iccmax=55A OCP>75A
TDC=30A Iccmax=60.1A OCP>80A
Vref=2V
FUVF:Vfb=0.2V
SUVP:Vcomp=3V
OVP:Vfb=2V
Fsw=320KHz

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POWER

U42@ PR3423 13.7K $\pm 1\%$ 0402 SD03413728J	U42@ PR3424 22.1K $\pm 1\%$ 0402 SD03422128J	U42@ PR3422 14K $\pm 1\%$ 0402 SD03414028J	U42@ PR3439 28K $\pm 1\%$ 0402 SD000015J00	U42@ PR3426 86.6K $\pm 1\%$ 0402 SD00001PM00	U42@ PR3432 24.9K $\pm 1\%$ 0402 SD03424928J	U42@ PR3434 1.18K $\pm 1\%$ 0402 SD00001D000	U42@ PR3438 29.4 $\pm 1\%$ 0402 SD00001BC00
U22@ PR3423 20.5K $\pm 1\%$ 0402 SD03420528J	U22@ PR3424 25.6K $\pm 1\%$ 0402 SD03425528J	U22@ PR3422 11.5K $\pm 1\%$ SD03411528J	U22@ PR3439 24.3K $\pm 1\%$ SD000011300	U22@ PR3426 130K $\pm 1\%$ 0402 SD03413038J	U22@ PR3432 22.6K $\pm 1\%$ 0402 SD000019W00	U22@ PR3434 1.1K $\pm 1\%$ 0402 SD03411018J	U22@ PR3438 40.2 $\pm 1\%$ 0402 SD034402A8J

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